To Blockchain or Not To Blockchain? An E-Voting Question

Shymaa M. Arafat
Associate Professor of Computer Science & Engineering
(Independent)
Final Draft that welcomes and encourages revising comments
Alexandria, Egypt
shar.academic@gmail.com, shymaa.arafat@gmail.com
ORCID:0009-0009-7571-4060, IEEE(CS): 102400508

***Abstract*—When blockchains appeared with its anonymous, tamper proof and immutable features, it became a tempting option for e-voting systems and countries welcomed many pilot studies around the year 2018. Then came a batch of opposing articles of which the most famous is an MIT paper in 2021 authored by a group of well-known security pioneers, joined by some blockchain experts, that describes the idea as "*Going from Bad to Worse*"; where they do not recommend internet voting, calling it bad to begin with and believe deploying blockchains will make it worse. Although most countries that participated in the pilot studies did not eventually integrate blockchains in their systems, many others like Russia, Romania, Mexico and some USA states did one way or another. As of today in 2026, literature holds thousands of papers proposing, or surveying, blockchain backed e-voting protocols/systems; many of which contain mixed information about deploying blockchains in the e-governmental system and using it in e-voting, about using it in private entities elections and using it in country level election, and about pilot studies or plans that were dropped at the end due to highly promoted media coverage at the beginning. In addition, to our knowledge, none of those papers defeated or discussed the arguments in the MIT paper; most of its citations came from blockchain related research or political legal research. Motivated by our own confusion as authors, this paper aims to fold the available material in four directions; 1) the opposing material with its argument, 2) the approving material with its defense, 3) the successful real world use cases at the governmental election level (i.e. separating from and/or resolving any mix-up with pilot studies and/or small/medium scale existing systems (companies/institutions/organizations) in the same country, 4) the intersection areas going back & forth between the two fields. This includes how e-voting systems benefited from the continuous innovations in Zero Knowledge Proofs (ZKPs) research accompanied different advancements in blockchain; also, how blockchain has innovated new vote buying/coercion techniques that take advantage of the anonymous automated online coding feature or the immutable public ledger feature. Finally, before concluding the paper, we investigate two existing blockchain-based e-voting systems and analyze them in light of existing criticism and/or compare with similar non-blockchain designs.**

Keywords—e-voting, blockchains, Zero Knowledge Proofs (ZKPs), vote coercion, vote buying, Decentralized Autonomous Organizations (DAOs).

I. Introduction

A simple search query on the two keywords "*e-voting*" and "*blockchain*" will lead to 4725 papers + 1 Book in Springer Nature, 3960 in ScienceDirect, 673 papers + 6 Books in IEEE Xplore and 25 in MDPI; this contains more than 600 review articles as can be deduced from Table I. Despite the rapidly increasing numbers, nearly 1600 papers up till now in 2026, they are not all directed to political country level voting systems. Especially during COVID-19 pandemic, online voting (with or without blockchains) became popular on all scales; hence, a lot of research articles proposed systems that are only applicable to small/medium-scale corperations (ex. [1/Table8]) which expands the scope of the mentioned numbers beyond the country-level case. Checking case-by-case in the two smallest numbers (25 from MDPI and 57 from IEEE in 2026), only 18 papers explicitly propose a nationwide system or a system/framework that can be applied at a large-scale country level. In earlier years, developed countries [2-5] dominate studies or proposals for a specific country, while many proposals were suggested for developing countries in the last few years [6-15]. As for review articles through the years, [4] is an early 2018 example, [16] is an 2022 example that discusses the later inclusion of sensor networks (IoTs) to capture physical data (whether fingerprint/biometric authentication or touch screen inputs at poll stations); finally [17] in 2026 tries to set a comprehensive framework when the field has matured enough.

TABLE I. SEARCH ON "E-VOTING BLOCKCHAIN"

Publication	*Total*	*2026*	*Review articles*		*Book*
			total	*in 2026*	
Springer	4725	773	158	40	1
ScienceDirect	3960	735	515	117	-
IEEE Xplore	673	57	>9	7	6
MDPI	25	4	5		-
Totals	**9383**	**1569**	**>687**	**164**	**7**

In general, there are two main approaches, [5], for deploying blockchains into e-voting systems. The first is to use a well-known public permissionless blockchain (ex. Ethereum [4/sec. 3.3, 5,7]) as an immutable anonymous public record with either a single vote or a batch of votes forming a transaction [18]; variants include hashes of votes, Non-Interactive Zero Knowledge Proofs (NIZKPs) or Succinct Non-interactive Arguments of Knowledge (SNARKs) as well. The second is to replace the usual Public Bulletin Board (PBB), [3/page 50], where election results are published at the end, usually by a private permissioned blockchain owned by election stake holders [4/sec.3.1&3.2, 16] or could be public in some designs though [5]; in that direction, researchers have proposed different consensus mechanisms (the validation criteria and the way in which participating nodes negotiate their decision) , like *Proof of Partition* [19] or *Proof of Authority* [13]. It is important to note that some designs deploy both private and public blockchains; for example Agora design since 2018 [20], Fig.1, that uses a private permissioned *Skipchain* (blockchain + skip lists [20/ref. 52]) to provide a faster consensus for ballots, then stores snapshot hashes of the

skipchain on the Bitcoin blockchain as a non-standard UTXO[1] (*Catena* schema [20/ref. 58]). In 2025/2026 newer designs (ex. DAVINCI [21], a July 2026 evolution of *Vocdoni*, in Fig.2) used Ethereum in a similar manner where ballots are stored in a custom Zero Knowledge Succinct Non-Interactive ARgument of Knowledge (zkSNARK) -based state machine layer2 chain on Ethereum (L2 ZK-rollup); *opn.vote* [22] in 2025, Fig.3, on the other hand used *Inter Planetary File System (IPFS)* (a peer-to-peer decentralized storage and file sharing protocol[2]) for actual ballot storage, while only metadata are stored inside the Voting Smart Contract (VSC) on Ethereum.

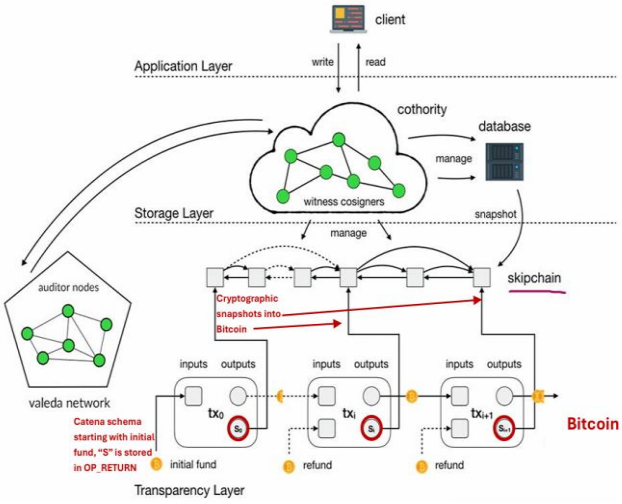


Fig. 1. Agora block diagram taken from [20] with some clarifying annotations added in red. Note that those Catena "*s*" values get stored in another list/chain Agora calls "*Cotena log*", where an entry (TX) is added based on consensus between nodes running Agora clients (see [20/sec. 3.1.2]).

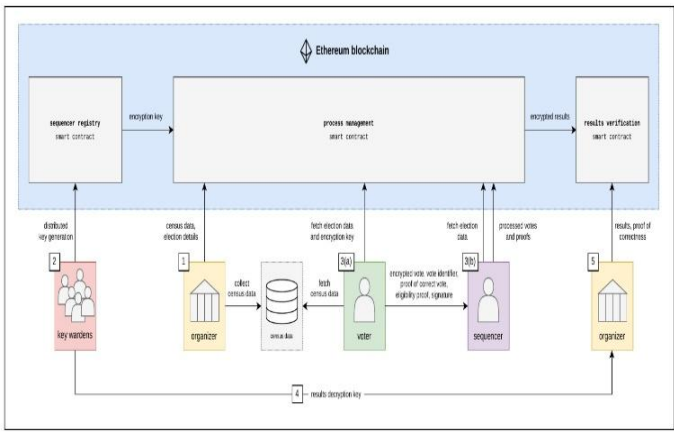


Fig. 2. DAVINCI protocol flow diagram taken from [21].

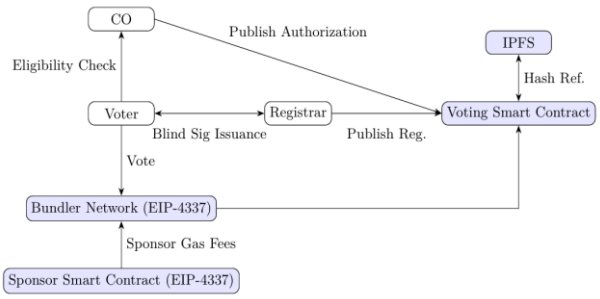


Fig. 3. Opn.vote high level architecture taken from [22].

Also, the reader should be aware that most blockchain voting design approaches can be suggested for remote online (internet) voting or poll station electronic voting using different kinds of voting machines. Lastly, the research direction discussing/testing public trust and/or acceptance to such systems, like [23-25], is beyond the scope of this paper.

Due to the huge existing material, this paper will follow a time chronological order; section II will summarize important pilot studies in the early years 2016-2018 and whether aborted or adopted, then section III will discuss the opposing research papers started appearing from 2018 up till 2021. Section IV will follow with other interactions between the two fields like new attacks that became possible with blockchains. Then, section V will discuss two case studies of officially used blockchains based e-voting systems (zk-voting in South Korea and Voatz with all the debates around it), while section VI concludes the paper. Finally, this paper assumes familiarity with basic blockchain and e-voting terminology; unfamiliar readers may find a preface and related terminology file in **[]**.

II. Experiments & Pilot Studies

Researchers in e-voting have long recognized a tradeoff between privacy (coercers cannot link counted votes to their true voters) and verifiability (voters should be able to verify the correct count of their votes to guarantee the integrity of the election). Blockchains seem to provide an easy way to achieve both; especially with the online coding capabilities added by Ethereum in 2015. Hence, many countries around the world started small-scale pilot studies, or funded research to test the water on the idea. This section summarizes important pilot studies and tracks the current status of their voting systems.

A. European Countries

In **Ukraine**, *e-Vox*, an NGO of developers who won the first Ukrainian contest of IT-projects on electronic democracy [26], developed and piloted probably the first blockchain e-voting system (on Ethereum) for city hall elections in Kiev since 2016 [27, 3/sec.4.2.3] and kept seeking to generalize the experiment up till 2021 [28]. **Switzerland**, with *Luxoft* company, performed an online voting experiment using blockchains (Hyperledger Fabric; private permissioned chain by IBM) on the city of Zug in 2018 [4/sec.3.1, 29,30]; 72

[1] Since Bitcoin allows only primitive scripts for coin transfer (does not have a smart contract capability), people used to piggyback their applications/chains to Bitcoin through what is called non-standard UTXOs; they write a regular money transfer TX, then put the value they care to post (usually cryptographic hash) in the OP_RETURN (unused otherwise) field. Catena schema starts with a BTC value *n* stored in a UTXO address *A* and each TX transfers a small amount from *A* to another UTXO; this can go on till *A* spends all the *n* value. In Agora design, "*Cotena log"* is another custom chain (append only list), called so because it contains a log of Catena values and a link to the corresponding Bitcoin TX; adding a new TX to *Cotena* is managed by those who have consensus rights on the election (nodes who are running Agora client software).

[2] https://ipfs.tech/; the paper behind it (https://arxiv.org/abs/1407.3561). **Account abstraction** (EIP-4337) is another important feature of *opn.vote* that allow voters to cast their votes (which involves posting ballot metadata on Ethereum) without paying/holding any cryptocurrency or even having crypto wallets to begin with; since this is irrelevant to our main discussion in this paper (no other mentioned system require voters to hold cryptocurrency), we will not discuss it in more details.

citizens out of 220 holding digital identities voted on things like whether fireworks should be allowed on festivals or not. Also in 2018, *Smartmatic* company offered to **Norway** a trial blockchain based online voting, *TIVI* [31,32], on a local referendum. Although *TIVI* (https://tivi.vote/) was born in **Estonia**, the country investigated then rejected [33] proposals to use blockchains in their long-standing internet voting system (since 2005, but the new design *IVXV*, which *Smartmatic* is part of, was first used in 2017).

Currently, blockchains may be used by the e-government in some European countries (ex. Estonia [34,35]) but not in general elections. However, the *European Blockchain Services Infrastructure (EBSI)* was first used by **Romania** to store (anchor) the cryptographic hashes of locally generated blocks, [36,37] in the first round of the presidential elections in November 2024; the system was designed by the Special Telecommunications Service (STS) of Romania (governmental design). STS maintained a private blockchain infrastructure for each election; the infrastructure utilizes the Geth implementation of Ethereum with a *Proof of Authority* (POA) consensus mechanism. In real time, information gets transmitted from this blockchain infrastructure to EBSI in the form of a transaction containing the block number and its hash; Fig.4. An earlier usage also happened, [38], as a voting reporting tool in the 2020 national parliamentary elections.

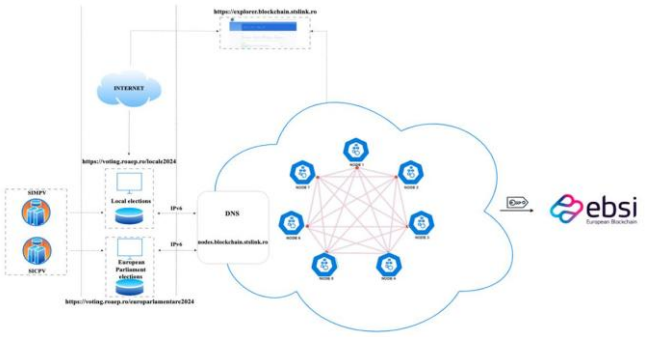


Fig. 4. Architecture of the System used in the Romanian Elections, adopted from [37].

On the other hand, countries like **Germany** [39,40] and **Norway** are still skeptical about the integrity of internet voting in general; while on the non-governmental level, blockchain e-voting usage existed there (and in Estonia and Spain). In Germany, there was a successful Ethereum-based online with 63,064 participant voters, *opn.vote* [22,41], that was deployed in a nationwide online self-organized referendum by **ABSTIMMUNG21** (a coalition initiative of civil-society organizations) in Oct 2025. Before that, *Smartmatic*'s 2023 online voting experiment (social election on health insurance fund) used blockchains (www.smartmatic.com/case-studies/, https://tivi.vote/experience/); also, the famous e-voting company working in Germany, *Polyas*, does not use blockchains [42]. Another success story is for *Vocdoni* from **Spain**, a design preceded DAVINCI, [21],that stores ballots in a private permissioned chain (*Vochain*) with BFT consensus; this was used by the opposition in **Belarus** [43] and by *Omnium Cultural*, a non-governmental organization in **Barcelona,** with over 180,000 members [44].

B. Russia

In Sep 2019, a pilot project conducted e-voting using a private blockchain "*Sovereign*" for the Moscow City Duma in 3 electoral districts [18,45], increased to 6 in 2021 [46]. The system was first tested on a university students council voting in July along with an international competition to hack it organized by the Civic Chamber whom Pierre Gaudry (Inria, France) won its half; [18,47] detail the system architecture, the earlier attacks and how they were handled. President Putin has officially signed the regulations for holding online voting for all elections and referenda in March 2022 [48], and Russia remained to expand the usage of Distributed Ledger Technology (DLT) in its internet voting system (known in Russian language as DEG [49,50]) through private permissioned blockchains, in spite of the criticism [46,51]. However, after the Sep 2025 elections, Russia declared a temporary rollback [52] of internet voting during the war.

C. Britain (UK)

Proposals in research articles can be traced back to 2018 [4,53]. A pilot experiment was conducted in April 2019 by a team from London School of Economics [54]; they used, [55], the open source of "*Verify my Vote*" with the voting system owned by Electoral Reform Services. Although UK government still funds flexible voting pilots [56], and research continues to further investigate the subject [57,58], online blockchain voting is still viewed as high-risk, [59], compared to traditional paper ballots.

D. Australia

Conducted some pilot studies around with a lot of promoting publicity, ex. Horizon State company in [60], but currently does not deploy internet voting in its nationwide elections [61/electronic voting]. To be accurate, sometimes voters have the option to vote electronically [62/alternative voting methods] in designated booths using Kiosk touch screens; the counting software, *eVACS* [63], is written in Ada programming language and does not use blockchains (available as opensource).

E. Africa: Sierra Leone

A Swiss company, Agora [20], ran a pilot project in the Mar 2018 presidential election that was reported in the literature as a proof of concept for the value blockchain e-voting [64,65]; Electoral staff manually recorded all votes of the western District into *Agora*'s blockchain platform to demonstrate its potential for transparency and fraud prevention. However, the company promoting it as the first ever blockchain elections [66] and also rushing to publish its results before the official ones [64], led to the government issuing a public denial, [67,68], clarifying that blockchain was not used in the official vote count and that *Agora* had no formal role in the election process.

F. Asia

In Sep 2018, **Japan** held a trial blockchain-based voting (for social contribution projects) in Tsukuba city [69], while **Thailand** main opposition party deployed the *Zcoin* blockchain (a privacy focused cryptocurrency that preceded

Zcash and rebranded to *Firo*.org in 2020)[3] in its Nov 2018 election with more than 120,000 members [70]. Although [4/sec.3.2], 2018, mentions a pilot in Bandung **Indonesia** using 2 private machines running the Ethereum protocol, we could not trace another credible source on that; [71] which is written by researchers from Bandung and published in an Indonesian journal did not mention such a thing in its survey part. Finally, **South Korea** started its first blockchain voting trial, [72], in Dec 2018 based on Hyperledger Fabric (private permissioned blockchain designed by IBM). Currently, we traced two important online voting companies in the country; *kevoting*, [73], not based on blockchains, used in **Kyrgyzstan** public office elections 2023 [73/2023 (04) & press (15,16)], and was about to be used in their 2026 parliament elections[4] but plans were cancelled [74]. On the other hand, *zkrypto* whose product *zkvoting* deploys blockchains and is officially used by the ***National Election Commission (NEC)***, [75], for public institutions elections and local community referenda [75/video]; however, the system is not used for public elections, [76], where poll station voting machines are used. Their system will be thoroughly discussed in section V.A (5.1).

G. Brazil

Brazil depends solely on poll stations electronic voting machines since 1996 up till now, with some criticism by security experts [77,78]. In 2020, the country requested proposals from technology companies to conduct a trial blockchain-based online voting with a sample of voters from three cities choosing fictitious candidates [79,80]. Also, in May 2025, a senator requested [81] to explicitly include blockchains and distributed ledgers among the technologies that may be considered for the purposes of studies and the development of new voting solutions.

H. Guatemala

In the 2nd round of 2023 presidential elections some Bitcoin developers from Guatemala civil society [82], in conjunction with Peter Todd a core Bitcoin developer, has used the Bitcoin Catena schema in a similar way to Agora (footnote1 and Fig.1). Peter Todd's *Open Timestamp* protocol [83] calculates a cryptographic hash of the submitted data along with the timestamp it was submitted to the protocol, then posts it to the Bitcoin blockchain; during the election, [84], timestamped fingerprints of partial results were each posted on the OP_RETURN field of a Bitcoin transaction. In more details, a *Merkle Tree* is built for each poll station results where node values are the hashes of a Ballot box count or a tally sheet scan; then, the *Merkle root* is considered as this poll station fingerprint at that time and is posted on Bitcoin along its timestamp. The process was promoted as "***Immutable Democracy***" [85], since Bitcoin was used as an immutable cryptographic witness on each Ballot box at a certain point of time against doubts of manipulations or altering in the aggregation process.

I. USA

As a first world country with the most advanced technologies, one of the earliest blockchain e-voting pilots, March 2016 [86], was conducted by *Smartmatic* company (*TIVI*); they ran a small trial in the **Utah** branch of the republican party voting for the presidential candidates; only usability issues were reported by voters. Seeming to be a supporting state for the technology, the first official blockchain vote (a semi-private permissioned Bulletin Board) was casted from Utah (using *Voatz*) in the October 2020 presidential elections [87]; the vote went to an independent cryptocurrency entrepreneur candidate.

Between the two incidents, 2018/2019, there were more than 30 blockchain-based e-voting pilots by *Voatz* [88] at the country level in three other states; West Virginia, Colorado, and Oregon. A $2.2 million funded pilot experiment took place in **West Virginia** primary elections 2018; the pilot targeted overseas military members, [89], where only 2 sample offices with couple of dozens of voters participated. Despite an attempted hack to West Virginia pilot [90], and wide concerns raised by academia on blockchain e-voting [86, 91-93], *Voatz* pilots were repeated for overseas citizens in **Colorado** Denver, Mar 2019 [88,94], then in **Oregon** two counties Jackson and Umatilla, Sep 2019 [95]. Through the years, researchers from MIT [96], and other places [97,98], got in a long battle against Voatz, but the fact is it is still used, [99], in many USA states and **Canada** (51 Ontario municipalities, but not federal elections) and the **Mexican** elections as their official site, [100], acknowledges.

The too many rejecting voices of internet voting in general[5] [101-106], and blockchain based systems in specific [86, 91-93, 107-112], prevented allowing it as a general option for country residents in USA. However, there are two groups of special case citizens whom their vote privacy will be jeopardized otherwise; without *Voatz* or a similar app, they have to sign a waiver of their vote privacy rights when sending their votes by regular post office mail [89, 94]; this holds longer delays to the election results as well. The first group that was included in almost all pilots is overseas citizens, especially military personnel (and their families) who are abroad in duty call; they vote according to the federal ***Uniformed and Overseas Citizens Absentee Voting Act (UOCAVA)***, [113], and the *Defense Advanced Research Projects Agency (DARPA)* is continuously sponsoring research for more secure options[6]. The second group is people

[3] Zcash (launched 2016) started in 2014 (with the name Zerocash) as an improvement on Zcoin to hide also the transaction destination (would be the Election Authority EA in e-voting) and the payment amount; this will be further discussed in detail in section 4 on nullifiers. For now, the interested reader can check this 2017 article (https://medium.com/the-zcoin-digest/zcoin-compared-to-zcash-4ad21d97636f); in 2025 (https://phemex.com/academy/what-is-firo-firo-guide).

[4] This link (https://qazinform.com/news/online-voting-planned-for-kyrgyzstans-next-parliamentary-elections-a82307) precedes the cited one in [74] by 2 weeks.

[5] According to [92], a famous quote of Ronald Rivest (one of the designers of RSA) is "*Best practices for internet voting are like best practices for drunk driving- there is no safe way to do either*".

[6] Read this preprint (https://arxiv.org/abs/2411.11796) rejecting a DARPA funded protocol proposed for remote voting, then read the acknowledgement in the updated version of the same MERGE protocol (https://link.springer.com/chapter/10.1007/978-3-032-05036-6_1) thanking the authors for their criticism that led to significant improvement of the final published version. The protocol suggests counting electronic votes using their protocol to save time, then apply a Risk Limiting Audit (RLA) methodology when the post office mail votes arrive to check if there is a significant error. This limits the delay cases to those when RLA flags an error.

with disabilities who state governments first faced some resistance when expanding *Voatz* usage to them in West Virginia [114], but then was included. As of now, Aug 2026, the law dictates a digital alternative (could be mail, Fax, or a web portal like Voatz) for those two groups, [115/internetvoting/], in 32 states and Washington DC, while internet voting is completely banned in 18 states (ex.: Georgia, Ohio, Illinois, Maryland, Minnesota, New Hampshire, New York, Wisconsin). It was worth mentioning in this context that there was another blockchain-based e-voting company *Votem* (https://votem.com/) that conducted some pilots in the years 2016-2019; its current platform *CastIron*, is used by both remote voting groups in Montana and by UOCAVA in Alabama [115/election-system/votem-castiron/].

However, some steps have been taken recently in the banning states; following are some examples that targeted the general election population. In April 2025, **New York** Assembly member Clyde Vanel (chair of subcommittee on Internet and New Technology) reintroduced a newer version of his proposed *Blockchain for Election Integrity Act* [116,117], which he tried to pass four times before; the bill, [118] proposed to conduct a formal thorough study by experts on how blockchain could help secure voter records and improve transparency. According to the official NY senate site [118], the bill is still active and has moved through the proper channels reaching the state senate on Jan 2026.

In **Georgia** [119] on the other hand, blockchain technology paved its way to Nov 2024 presidential elections; Screven county applied the same technique that was applied in Guatemala through *SimpleProof*, [120], (an election specific startup founded by the same Peter Todd in [83]) to post a timestamped cryptographic proof of the poll station election results (as taken from the air gapped voting device in the poll station) on the Bitcoin blockchain. In this technique, blockchain technology just adds an extra security layer without changing the existing systems or disrupting their functionality. Although banning internet voting in general [115], this was more than justified for Georgia after previous problems, [121], that happened in 2020 elections regarding some partial results (certain voting machines); other similar problems [122], for example the Wisconsin 2024 recount (due to suspected tampering of records from a certain machine for leaving its door open during night), could have been replaced by calculating the cryptographic fingerprint of the current data and comparing it to the one posted on Bitcoin[7].

Note however, that this technique does not prove the integrity of the results before the time it was hashed; it could have a reverse effect of asserting fraud data if data was somehow manipulated before the proof is generated. Josh Benaloh described it in [86] as taking a picture of an existing picture, we say yes it is not like a picture from the actual scene, but it is much better as a proof. If a losing candidate took a photocopy or a mobile photo of the subtotals counting sheets, the Election Authority (EA) and/or the winning candidate could claim in court that the photos are fake and the court has no way of knowing (beyond reasonable doubt) which sheets are fake; while, if the cryptographic fingerprint of the sheets is on a public trusted blockchain, we could think of the blockchain as a trusted as an analogy to a "never-lie & never-forget" hearsay witness that saw the actual true sheets. Even if the EA tried to change the finger print before being posted on the blockchain[8], a photocopy or mobile copy with their digital signature (using the EA secret key) cannot be claimed fake with overwhelming probability.

J. Mexico

As will be discussed in section 5.2 (V.B), Voatz successfully conducted online voting in the Mexican 2024 election for nearly 144,734 citizens; 234,117 registered to vote remotely then 122,496 voted through their mobile phones and 22,238 in person at kiosks with their votes still handled by Voatz[9]. This special case election had contests for President and Senate, and some states had contests for Governor and/or another local office, all in one ballot; 428,049 contest selections in total.

III. Voices Rejecting Deploying Blockchains in Voting Systems

In this section, we will go through the main arguments of all researchers who wrote against the use of blockchain in e-voting through the years, [86, 91-93, 107-112], to discuss which of those arguments are still valid and in which cases. However, we will skip points against online voting in general since this is not the scope of the paper; there are many cases when blockchains are used with poll station voting as was discussed in section 2.

A. Controlling the Chain (Public Permissionless blockchains)

The main doubt raised by Josh Benaloh in [86] is that when it comes to major elections, we cannot just trust anonymous miners and/or full nodes to not add or alter transactions. This includes the commonly circulated risk of an adversary dedicating huge resources to attack the network. The possibility he mentioned of miners having some biases or conflict of interests towards a certain outcome (ex. voting on a candidate calling for a high tax on miners) can be generalized as the case of colluding/malicious miners or node operators in general; the cost and handling of the attack here will be different since those already have running nodes (no setup cost involved). In what follows we will examine the feasibility of each of these threats, but first we need to preface with some related information. Through the analysis, some issues raised by [91,93,101,107,111] will come up and hence will be also discussed.

1) Preface

[7] Credit of the Wisconsin example belongs to Chain-link educational hub (https://chain.link/education-hub/blockchain-voting-for-elections), which reminds of the fact that the general idea of this solution is applicable on any trusted public permissionless chain (like Ethereum in Vocdoni design in [21]) not just Bitcoin.

[8] We mean replaced the TX with another one with a higher fee to get minted first (RBF in Bitcoin when OP_RETURN method is used like here; or for any other blockchain, simply putting another TX in the same pool with much higher fee to get picked first and make the original one look like a double spending attempt). If this ever happened, **the attendants still have a copy with the EA digital signature;** note however, that this **is not applicable to Guatemala case** since the process is solely conducted by civil society, while in Georgia it is done by the election officials.

[9] As an indication to system scalability and design challenges, the authors from Voatz say in [182] that they had to upload the records of ~100 million eligible voters into their servers (not just those who registered to vote remotely) and decrypted all the 144,734 ballots (including those who voted at kiosks).

Let us set the ground with some related definitions, numbers and rationales.

Computational Security:
When analyzing systems security, the term "*Computationally Secure*" considers the system secure if attacking the system is computationally infeasible with overwhelming probability; i.e., it takes a lot of money, time, and/or computation power to penetrate the system than an adversary could possibly have or find it worth the gain from the attack. Specifically, in nation level e-voting systems, the amount of resources (money and computing power) an adversary may dedicate to an attack depends on whether it is just a party competing to win, or a powerful country that has farsighted interests in the elections result; naturally depends too on the country and election type. The adversary's budget will differ whether it is a local parliament election, a presidential country election and which country, or a European parliament election in case of European countries; also, whether it is a peace or war time (the most expensive cyberattack maybe cheaper than war physical weapons)[10]. Having that said, let us look at the numbers.

Numbers:
-In USA, where most of the rejecting voices came from, the average presidential campaign funding for one candidate [111/ref.48, 123] was roughly $768 million in 2016, jumped in 2024 to ~ $2 billion by candidate and a total of $8 billion raised by the campaign. Agreeing with [111], this means that the adversary budget to disrupt or change the result of such election should be estimated to reach that number.
-The cost of possessing at least 51% of Bitcoin mining power was estimated by a study from Duke University Business School (the same professor works also in National Bureau of Economic Research in Cambridge) in 2025 , [124], to be at least $6.07 billion ($4.6b for devices + $1.47b for electricity and maintenance), assuming the availability of the needed electricity and mining devices; newer 2026 rough estimates raise this number to $8 billion [125] and may have increased further, [126], with the increase in energy prices due to rising wars.
-The corresponding threshold for most Proof of Stake (POS) chains, including Ethereum since Sep 2022[11], depends on the total staked currency; although most existing POS chains do better than that, the theoretical lower bound is more than 1/3 of staked amount. As of Sep 2026, [127], total staked Ethereum is 42.632959 million ETH ~ $105 billion which means an adversary must first stake more than $35 billion worth of ETH to think of controlling the network. Here, the cost of running the needed nodes (validators and block builder/proposer nodes) is far less than Bitcoin and can be ignored compared to the staked value.
-Also interesting are the numbers (ratios) of power distribution between pools as captured from live monitors (Sep 2026). In Bitcoin, [128], the 10 days average shows Foundry USA pool mined 26.087% (390 blocks) followed by AntPool with 17.726% (265 blocks); note that these monitors are approximate or rough estimates, as they cannot capture existing hash power that failed the competition and did not mint a block. In Ethereum on the other hand, where people can stake smaller shares through large staking pools [129/pools/], [127], the largest pool Lido holds 21.49% (decreased from 27.94% on a prior check at 14/1/2025) followed by Binance with 7.6%.
-As for geographical distribution, live monitors on Bitcoin [130] shows 31.6% active full nodes in USA followed by 14.85% in Germany (~ 40% in Europe) in Sep 2026; while a validator deanonymization paper in 2025, [131/Fig.9] discovered 46.79% peers in Europe and 33.01% in USA.

Decentralization:
Although decentralization is an essential main feature in blockchain design, the ability to participate and the degree of participation still depends on the ability to afford resources. Hence, researchers have long studied the degree of decentralization for each chain through different performance metrics; [132] is a Systemization of Knowledge (SoK) 2025 paper that summarizes previous research, then studies different metrics on different blockchains and provides an interactive live dashboard for them [133]. Also, [131] captured another interesting angle when discovered that many validators in Ethereum (could similarly be traced for Bitcoin miners) distribute their power among a number of pools creating dependencies between pools and discusses the implications of that [131/sec.8.4] on the degree of decentralization; i.e., there is another dimension and the above ratios in live monitors are not enough alone.)

Censorship Resistance & The Anonymity Property
Anonymity in blockchains is implemented via what is called Public Key Infrastructure (PKI) where each user generates a private-public key pair; users sign transactions with their secret private key and others verify the signatures using the public key. Thus, users are identified only in the transaction through their public key address, and in general users are advised to generate a new pair for each transaction. Hence, one should expect a first line defense for election authorities is to keep hiding from adversaries through a chain of addresses and keep generating new ones for every step. However, things are not that easy; even if election applications did more than that and tried to use encrypted transactions and/or hide using things like Transaction mixers and Tor applications; challenges remain high, [134], and the IP address generating the transaction can be identified in most cases. Another approach is to use privacy focused blockchains like Monero and Zcash [135]. Still, real life examples of this approach can be found in crypto crime reports, [136], of criminals and sanctioned countries who tried to do that; yet, tracing remains possible especially with higher adversary powers.

2) Bitcoin
1-The fact is, up till now, double spending attacks have happened, [126], in other smaller yet well-known cryptocurrencies' blockchains like Monero (2025) and twice in Bitcoin Gold (2018 & 2020) and Ethereum Classic (2019 & 2020); however, it never happened in Bitcoin as a much larger chain with much higher and more solid barriers.

a) If an adversary invested all this money to perform a double spending attack or to reverse the last unconfirmed 6 blocks (cannot reverse or alter earlier than that), the money gains (in Bitcoin currency BTC) will be almost negligible to the cost incurred; this will also shake trust in the currency making the gained value decay further. The author in [124/ sec.

[10] Recall in section II.B (2.2) Russia suspending its e-voting system due to war [52].

[11] Before Sep 2022, Ethereum was Proof of Work (POW) chain like Bitcoin; the reader should put in mind this fact when considering any estimated numbers or written arguments before that date.

4.A] argues that the adversary could invest in perpetual/conventional futures that rise when BTC price goes down; the author then added that nefarious actors might target the network for other reasons[12], which exactly describes the profile of election adversaries.

b) A defending argument is that attacking Bitcoin is not just about money, the needed electricity is a *huge physical barrier*; even at the nation level, a country (say Russia/USA) gathering the needed electricity can hardly go unnoticed by major intelligence agencies of other countries (say USA/Russia respectively). Although the existence of rentable hash markets "*Nicehash*" (https://www.nicehash.com) which may facilitate a hit-and-run attack, it clarifies that the "*Nicehashable*" metric does not exceed 5%[13] in Bitcoin; hence, it is not enough to break the physical barrier. However, **what if an adversary distributed the mining devices (and hence the needed electricity) into say 20 different nodes in 20 different countries, will gathering the electricity be as noticeable by intelligence agencies?** We believe the answer remains debatable.

3- For the possibility of biased miners or mining pools, things get more risky as the live ratios of active full nodes, [130], and pools [128] reveal; the threat increases with newer research findings. For example, the authors in [137] use a game theoretic model to argue that a probabilistic double spending is possible with controlling only 26%; this, if applicable, can half the adversary's budget, what if combined with bribing miners and rentable hash. Actually, it seems a feasible scenario to imagine a competitor or an adversary in a USA presidential election achieving the 26% barrier (or even 51%) by splitting his/her budget between buying people and buying devices with some renting if possible. Although not that feasible for small developing countries that may benefit from the idea of immutable democracy, opposing voices may doubt foreign interference from USA or Europe who control a significant ratio of mining pools and active full nodes. However, a defense maybe that those claims already exist with or without the blockchain.

4- As for USA and similar powerful countries, it is expected for every concerned entity to run a Bitcoin full node in that case leading to attack detection where the truth remains a consensus between those entities. Governments can go a step further and run a significant ratio of nodes during election time; this was described by Ben Adida in 2017, [107], in the case of handling just meta data (like fingerprints) as solving a small part of the problem with an *unnecessarily big hammer* and by Nir Kshetri in 2019, [93], for the general case as unnecessarily exhausting the government budget.

3) Ethereum and POS Blockchains

1-In addition to a starting budget of more than 1/3 of the total stake ($35 billion in case of Ethereum [127]), POS chains deploy extra techniques, [138], to both increase the 1/3 threshold and to limit its attack effect. First, by detecting and slashing the stakes of malicious nodes (those who validate conflicting blocks) in the next epoch. Meaning the adversary here will lose a significant portion of his/her stake if the attack failed; this raises the threshold to 5/9, [139], of total stake for the attack to keep going after the first slash. Meaning that the loss after a failed attack will be more significant if compared to Bitcoin case (POW in general) where a failed adversary will lose only the electricity and maintenance cost (~ $2 billion) but still have the mining devices (~$5-6 billion) to keep or sell.

2-In addition, recent studies, [140,141], show that higher lower bounds can be achieved in reality when considering which staking nodes (accounts/addresses) are currently active and participating in the block proposing-building-validation process; this can dramatically raise the lower bound to 99% of active stake in some cases. Considering the slashing mentioned in 1, the attack resistance may increase further by calling honest "*sleepy*" clients as [141] describes them to actively participate after discovering a malicious address and slashing a portion of its stake. In that direction, Ethereum also introduces some penalties for unparticipating validators (those who miss their turn to post a block when selected as a leader); if (>1/3) of the validators went offline, this can cause a threat to finality and Ethereum impose "*inactivity leaks*" penalty for that. Comparing to Bitcoin (POW), governmental participation here will be through staking and maintaining continuously online validator nodes with a much less cost than that of running Bitcoin full nodes; i.e., the invested budget will remain (only subject to ETH price changes) and may even result in some added profit.

3-As for existing validators/pools who may have a conflict of interest (certain bias towards the election results), or the possibility of bribing them, we consider two threats. If they tried to jeopardize the liveness of the network (does not post a block when chosen as a leader) or to validate contradicting blocks/transactions, they will be penalized of slashed respectively; in the contradicted validation case, they may be traced as criminals and become a subject of trial if proven to do it deliberately for disrupt the election course. However, validators can still post other blocks that do not contain all or some election transactions (to delay all in general or to skip some specific transactions like certain ballots or sub-results), or even worse manipulate the transaction without being discovered if kept enough majority for a few epochs (colluding with the EA to change the partial result as discussed in footnote 8). We have already mentioned in [131, 132] that concerns are already discussed on the literature regarding the degree of decentralization and concentration of power; [142] is a newer 2026 study on the risks of staking pools when there are malicious agents.

4-On the positive side, most POS chains invest a lot of effort and money in software auditing and in making the leader choice of a certain block purely random [143]; meaning that the adversary's chance in controlling a certain block at a specific point of time (an election fingerprint posting for example) remains probabilistic and pretty much unlikely for a same cluster to control consecutive epochs. In addition, Ethereum is a continuously evolving network that keep fixing exploits; for example, the authors of [131] declare that they

[12] Note also that, unlike other blockchains that allow smart contracts with programming capabilities, Bitcoin allows only a limited number of scripts (TX format). Consequently, this limited its usage to mostly money transfer TXs; i.e., haven't faced many application cases where hackers' targeted gains are not just financial.

[13] Although the live ratio on (https://www.crypto51.app/) is set to 0%, the (https://www.kucoin.com/blog/en-understanding-the-moat-what-is-pow-51-attack-cost-in-2026) which was last accessed 7/9/2026 but is not available says it is 5%; beside the fact that the Nicehash site is running and active mainly as a Bitcoin hash market.

received a bug bounty for their findings and the ratio of the leading pool "Lido" decreased ~ 6.5% last year.

4) Concluding Remarks

In this subsection we have gone through the two major chains Bitcoin & Ethereum; also cited a proposed research article that uses a privacy focused chain. The reader should be aware now that monopoly threats do exist, but still have a defense and/or protection strategies that makes them debatable most of the time. Countries and involved entities are expected, and advised, to invest some money (stake/hash rate) in the used blockchain to gain a safety margin power threshold. Finally, it remains a different decision for each country to estimate the power of its adversaries and weighs the pros & cons of each choice. The scientists in USA for example fear money power intervention the most because estimated adversary's power is almost the highest worldwide, and see paper ballots as a safe option, [111], because they are not faced with problems like those reported in the video about Guatemala [85]; i.e., there is no single recipe that fits all countries.

B. Controlling Private Permissioned Chains

Although this section is mainly about the threat of controlling major large-scale public blockchains like Bitcoin & Ethereum, we point out that the monopoly risk is naturally more feasible in private permissioned chains. When a custom private blockchain is designed to take the place of the traditional public bulletin board (PBB), it usually consists of a limited number of servers that validate transactions according to some consensus rule. Featured examples include zkVoting (South Korea) [75] that runs Ethereum protocols on a local Ganache environment and hence consensus (the cryptographic rules on which transactions are validated) is done according to a quorum of trustees. Also, Voatz (USA, Canada, Mexico) [96/sec.5, 97/Blockchain servers] that uses 32 Linux servers split equally between Amazon AWS and Microsoft's Azure; hence, it remains vulnerable to possible control by Microsoft or Amazon if not by surreptitiously adding resources as mentioned in [96,97] could be through trapdoors or exploits in the servers or their operating system[14].

C. Software Dependency (Attacks &Vulnerabilities)

We start with some software related terminology definitions, two of which are included by the MIT paper [111] in the 5 minimum necessary but not sufficient requirements for secure elections in an evidence-based framework; ballot secrecy, software independence, voter-verifiable ballots, contestability and auditing. Then, we move on to discuss the software related concerns (code vulnerabilities) and the defending arguments against them before deducing any conclusion/recommendation at the end.

1) Terminology & Definitions

Software Independence [111, 111/ref.35,36] is defined as *the property that an undetected change or error in a system's software cannot cause an undetectable change in the election outcome*.

The definition clearly contains software attacks & vulnerabilities and extends to include any deliberate code alteration say of an opensource election code for example. Whenever there is a dispute about the existence of an error/alteration, another metric is defined for elections.

Contestability [111,111/ref.37] *A contestable voting system is one that provides publicly verifiable evidence that the election outcome is untrustworthy, whenever an error is detected.*

This property relates to providing a publicly verifiable audit trail to the election results; a cryptographic proof (usually a zero knowledge one to keep vote secrecy) that the submitted list of ballots is the same counted list of ballots.

Transparency [111] Transparency in voting systems usually refers to auditing the election process in all its steps by candidates' representatives and neutral observers (civil society, local and international NGOs, ...). However, in electronic voting systems, transparency also includes *disclosing the system design, the cryptographic protocols used and open sourcing the election code (with documentation)*.

-Many existing systems lacked such property that researchers had to apply reverse engineering on them to get some clues; examples include the Russian voting system [47,144] and Voatz in USA [92,96-98]. In fact, we believe most blockchain rejecting voices in USA were originally motivated by their suspicion due to the secrecy of Voatz design; still after reverse engineering, [96], the authors could not determine what is exactly stored in the blockchain and had to optimistically assume that all cryptographic activities are logged to the blockchain, and that these blockchain records are secure, monitored, and immutable. Recall that the title of [96] clearly states that "*The Ballot is Busted Before the Blockchain*" and the abstract describes their work as a "*concrete illustration of the common wisdom against internet voting and* ***the importance of transparency***".

-Also, a related issue here is to prove that the open-sourced code (if existed, hence analyzed and approved by involved entities) is the one actually used during the election (i.e. has not been altered).

2) Concerns

1-The scientists and associations in 2018 [91,101] were concerned about the fact that blockchains like any software can have exploits and live a continuous evolving battle; from cryptographic attacks to design to simple software exploits when thousands of code lines are written. When people say we use it every day in our financial transactions [?], the MIT paper in 2021 [111] replies that attacks do happen in financial cryptocurrency transactions and even in digital banking systems, so why risk it in elections when the adversary' power could be much larger and the chance of tracing the thief or retrieving what was stolen (your vote) is far less in elections. In this context, we bring up a famous 2016 incident of *Ethereum reorg*, [145/DAO fork,] to return more than 3.6 million ETH that was stolen in a hack[15] by withdrawing the money from the original smart contract (transaction) before

[14] This is an equivalent threat to buying Ballot Marking Devices (BMDs) manufactured in another country; watch this 2017 lecture by Dan Wallach (https://youtu.be/CI8q_356zZs) where he discusses possible threats even if the BMD is isolated.

[15] There was a voting on that with 85% approving (https://blog.ethereum.org/2016/07/20/hard-fork-completed); those who objected insisting on keeping the principle no matter what (Code is the Law) forked into another cryptocurrency (Ethereum Classic).

the hack time; most election fraud scenarios will not probably need a *reorg*, but it might be harder to reach consensus on which transactions are fraud and which are the correct ones.
2-The paper in [111], and the US Vote foundation in [112], point out to another risk about **the attack scale**. While an alteration of paper ballot boxes is physically noticeable and limited in size, an alteration of a few lines of code could possibly change millions of ballots and is harder to detect (recall the related issue to transparency).
3-The authors argue that software independence should be **checkable using non-software means**; they compare to paper ballots being physically available for recount (comparing the strength of a hardcopy vs. softcopy document as an evidence/proof).
4-An attacker willing to spend enough money could flood the blockchain with transactions, [111,112] to drive up fees and keep users from voting until after the deadline passes.

3) Defending Arguments

1-Of course, Ethereum, and most smart contract enabled blockchains that have an Ethereum Virtual Machine EVM design, are more susceptible to newly discovered attacks or software vulnerabilities when compared to Bitcoin primitive script language. For that reason, they endure continuous updates for improvements and/or fixes (hard forks [145]) unlike Bitcoin that rarely invokes such option.
2-In all significant blockchains, adding or updating any piece of code goes through many software auditing and testing phases before the actual integration; besides, it is all open-sourced. As stated earlier, many USA rejecting voices were partially influenced by the lack of transparency in the used system (Voatz); this does not have to be the case in other systems. In fact, open sourcing the code remains a debatable issue even in systems that do not use blockchains; for example, the voting application in the Estonian voting system [146/Assessed Threats/(Open Code, Closed Code)].
3-Cryptocurrency defenders think of major cryptocurrency blockchains like Bitcoin and Ethereum as having the "*Too Big to Fail*" property. They hold a lot of applications, a lot of money transactions and a large market cap; attacks/exploits can be rapidly discovered and traced, [136], even if sometimes it becomes hard to return the stolen money. Election-wise, when there is no money to retrieve, an attack or any kind of manipulation or network monopoly can hardly go unnoticed; i.e., like any other voting mechanism, a worst-case scenario is a recount or re-election. In fact, especially in hybrid systems when used with poll station voting as in Georgia [119], they may better help in locating the error location (identifying the faulty partial sums) to minimize the recount efforts (the examples from previous elections in [121,122]).
4- Specifically for election auditability and contestability, almost all online voting systems, including those who deploy blockchains, include a publicly **verifiable audit trail** using NIZKPs (Non-Interactive Zero Knowledge Proofs). Many of them, and also many in-booth e-voting systems, apply some kind of physical checking of a random sample of votes using statistical error measuring techniques; i.e., **Risk Limiting Audits (RLAs)** [115/audits/whatisrla/, 147].
5- An important argument that we partially disagree with [111], is about pen-marked paper ballots. We believe paper ballots are not necessarily more **voter verifiable** and/or coercion resistant; yes, a hand marked paper ballot is unbeatable evidence on "*cast & recorded as intended*" but scores much less than electronic voting in "*counted as recorded*". There are many reported cases in many countries (ex. Guatemala in [85]) where complete ballot boxes could be altered. Regarding **coercion resistance** and **ballot privacy**, malicious poll workers or crowded poll stations may jeopardize vote privacy and other aspects as well. In addition, if coercers could get just one empty paper ballot, there is a technique called "*circulating ballot*" where coercers/vote buyers hand the voter an already marked ballot as they desire to throw in the ballot box and require him/her to give them the new empty paper ballot to repeat; the best the coerced voter can do here is destroy the marked ballot and not vote at all. We agree that both internet voting and blockchains add new attack vectors (especially in vote buying/coercion as will be discussed in section), but we also point out that pen marked paper ballots have their major issues too.
6-We believe e-voting systems featuring those posting cryptographic fingerprints on the blockchain like [20-22,83,85] scores much higher on **contestability**. No voter could ever verify that his/her pen-marked paper ballot is included in the final count; how would a losing candidate (or election observer) prove that the partial counts he/she witnessed led to different values than the published ones when draft paper photocopies are not enough as a proof in court. For those cases, most internet voting systems provide QR code proofs which may not be enough evidence in case of dispute, while blockchain-based ones provide immutable publicly available proofs; you may doubt those software cryptographic proofs and/or worry about the hidden exploits, but you do not have an equivalent option.

4) Concluding Remarks

We believe the concerns raised by academia and associations do not include the case of just adding an extra security layer to the existing systems through posting cryptographic fingerprints of partial results on the chain [85,119,120]; the authors of [111] state clearly that they are focused on internet voting and that using blockchains to help in other aspects of election like auditing is not covered in their paper. Then, we again argue that it is a case-by-case basis decision for each country to weigh the pros & cons. In general, we recommend posting partial timestamped fingerprints (the *SimpleProof* approach [120]) for countries that use poll station voting. Especially those who use pen marked paper ballots (no voting devices), this can provide a significant security improvement with minimal cost[16]; careful timing should be chosen to prevent prior alteration, and the software calculating the cryptography should be opensource and audited by involved entities. If done right, with enough auditing to make sure no manipulation was done earlier, this can make any voting system scores better on verifiability and contestability. Also, we recommend complete transparency in e-voting systems design; the architecture, the protocols and the code.

[16] even in Germany as a developed country, the cost reduction achieved by *opn.vote* pilot [22,41] was highlighted as one of the main advantages resulting from using Ethereum-based online voting (reducing the costs per vote from €2 to under €0.01).

D. Coercion & Vote Buying

We have elaborated in the previous subsection that pen-marked paper voting systems have their own coercion and/or vote buying risks that are not possible in online voting; even if the coercer is living with the voter as [111] says, the odds of sneaking a moment away or with different device remain higher than in the fixed short time in poll stations. However, we did not deny though that a new set of vulnerabilities come with online voting and another may result in case of using blockchains. This section summarizes those threats raised by [92,111,112].

1) Threats that are general to most verifiable e-voting systems

1-A straight forward threat for all internet voting systems is compromised devices or compromised communication channels; [111/systems attacks] discusses that threat in detail with real world examples. Existing i-voting systems like those in Estonia, Canada, France, and Switzerland have it in their regulations that a compromised voter device must not be enough to forge a vote; however, we are here concerned about revealing the true vote to the vote buyer/coercer.

2-In addition, whenever a receipt or a QR code is provided for voters to verify their votes, it can be used by vote buyers and/or coercers as well. Many e-voting systems mitigated this dilemma by allowing multiple voting [148]; you can simply show the coercer the old receipt/QR code then vote again as you wish. This can be defeated if the coercer can find out somehow all your voting transactions or the last one (or its timing) or can force you at the last final moments before voting is closed.

3-Section 4 (IV) will discuss new vote buying/coercion threats that resulted from the existence of blockchains, even for e-voting systems that do not use blockchains.

2) Threats emerging from blockchains deployment

-In 2018 Ari Juels and his colleagues, [92], expressed concerns that blockchains make it easier for vote buyers; an experience they realized from cryptocurrency related small scale e-voting in Decentralized Autonomous Organizations (DAOs) through governance tokens.

-At that time, they were working on an idea about vote buying threats they called DARK DAOs [149]; they showed how conditional payment to the vote seller (automated via a smart contract to his/her wallet) can be triggered by a remote execution attest from a Trusted Execution Environment (TEE) that a certain code has been executed[17]. So, if a certain e-voting system assigns voting rights to voters as tokens or coins, [111/coins as votes], the existence of Trusted Execution Environments TEEs, 2019 [150], on the voter device may facilitate buying or tracing that token[18].

-The research group pursued the thread further in the recent years, 2025 [151-153], to demonstrate how a token or its access rights can be rented secretly for a fixed interval (the election days) which facilitates vote buying. Of course, this threat does not exist with poll station voting whether using pen-marked ballots or using voting machines when posting cryptographic fingerprints on a blockchain (no tokens or TEEs are involved).

E. Malleability Threat

1) Preface

-Malleability as a property is mainly related to the chain deployed cryptography; if two different plaintexts can result in the same cryptographic value (hash/digital signature/encryption/….), or can have more than one valid cryptographic value, then an adversary can replace one with another. Many e-voting papers, [39,154], either prove or start by assuming that all cryptographic protocols used are Non-Malleable against Chosen Plaintext Attacks (NM-CPA); the most common approach (goes without saying when using shuffling mix-nets for example) is to use NIZKPs [155]. Blockchains also have their malleability problems that may affect the e-voting systems using them.

Transaction malleability:

-Blockchains, and many e-voting systems, deploy Elliptic Curve Cryptography, where cryptographic functions can be equivalent to calculating a point on the elliptic curve used.

-It was first noticed by **Bitcoin**[19] , [156], that the mirror point on the other side of the axis could be considered also as a valid signature. Putting it in the simplest possible way, since calculations are all done modulo a very large number say N, any given signature can be represented as (R, S) or (R, N-S); this makes it possible to change the transaction ID of a confirmed transaction which may be used to steal coins. Finally, in 2017 the *Segregated Witness (SegWit)* prevented Bitcoin transaction malleability, among other enhancements like scalability regarding the block size, when it separated all transactions signatures (usually called cryptographic witnesses) at the end of the Block.

-As explained, the solution is not general to all blockchains; for example, the **XRP** ledger (**Ripple**) [157] completely solved malleability for single-signed transactions in 2020, while threshold multi-signature transactions can still have malleability issues if they have more signatures than the necessary threshold (M-of-N threshold homomorphic signature).

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[17] Remote execution attests were originally discussed on Intel SGX (which is available on many new PCs in the market: https://www.intel.com/content/www/us/en/architecture-and-technology/software-guard-extensions-processors.html), and it exists in other processor companies like Apple and others as well. So, we can assume there's a considerable probability that the voter PC supports it.

[18] Vitalik Buterin et al. introduced a new notion, **"CK" *Complete Knowledge***; as opposed to previously existing Proof-of-Knowledge protocols, a proof (or argument) of CK shows that a prover not just knows the targeted secret but also has fully unrestricted ability to use it (*Complete Knowledge: Preventing Encumbrance of Cryptographic Secrets";* https://eprint.iacr.org/2023/044). However, this won't help if the voter device is totally compromised (the hacker has fully unencumbered knowledge), if the voter is voluntarily/coerced selling his/her vote or if deploy the techniques in [153] to rent complete usage.

[19] A subset of the general malleability problem was first solved in Feb 2013 released update, Bitcoin core 0.8.0, by accepting only new TXs with digital signatures following Distinguished Encoding Rules (DER) format, Feb 2015 core 0.10.0 made it strict (https://gnusha.org/pi/bitcoindev/55B79146.70309@gmail.com/); the point from the previous link about OpenSSL is for the reader to understand how fixing and/or discovering malleability cases can get quite technical and is specific to each chain. The interested reader can find the exact sequence of events and links to discussion traces in (https://en.bitcoin.it/wiki/Transaction_malleability#Signature_Malleability)

Merkle Tree Malleability:

-In addition to digital signatures, Merkle Trees are another widely used cryptographic construction, whether by blockchains or by e-voting systems even without blockchains. A merkle tree is built from all hashes (transactions in a blockchains' block or could be ballots in an e-voting system); the tree can be used to derive a cryptographic witness on the existence of a certain leaf node where the root value is included in the block header of most blockchains. We can describe merkle tree malleability as the ability of an adversary to alter the tree nodes in anyway while the Merkle root remains the same (ex. replace an intermediate node with a fraud leaf) without being detected; the authors in [158] (2024) discuss the security implications on blockchains in general.

-In **Bitcoin**, there was an old Merkle tree malleability, CVE-2012-2459; if the number of transactions is odd, the last value is duplicated which means that repeating the last transaction twice will lead to the same Merkle root value. This was fixed in Bitcoin core 0.6.1 2012, [159], by rejecting a transaction with a hash value already existing from inclusion in the block. Another one that could lead to deceiving users with fake inclusion proofs for their transactions (a transaction with exactly 64 bits of non-witness data can be interpreted both as an intermediate node in the tree and as a leaf in the tree); this was fixed recently in April 2025 by BIP 54 (https://bip54.org/).

-Another case example is a famous attack, [160], on a Merkle tree library widely used by developers in Ethereum, *OpenZeppelin*, where it was possible to construct Merkle trees that allow forging a multi-proof for an arbitrary set of leaves. Note that the vulnerability, discovered in 2023, is quite similar to what was fixed in Bitcoin BIP 54; meaning that such vulnerabilities may very well be invoked again in another blockchain or open-source library due to the code implementation method.

Concluding Remarks:

Most large-scale blockchains have mitigated their transaction malleability problems in a way specific to their transaction format and/or design. Hence, the threat remains significant only when customized blockchains are designed for the e-voting system without handling/checking the existence of such vulnerability or using the famous ones in a way that could revive the exploit. The threat exists also, for e-voting systems that use these cryptographic constructs even without blockchains either by mutating an individual ballot or a group of ballots; for example, a hypothetical e-voting system that depends on a Merkle tree of the ballots lists for proving votes integrity (individual and universal verifiability) should check malleability risks of the used Merkle tree implementation.

2) Transaction Malleability Threat to e-Voting systems

The general threat, as mentioned above, has many possible scenarios. These two papers, 2019/2021 [161,162], conducted simulation and testbed experiments concerning transaction malleability in e-voting applications. The authors created a mutated malleable transaction T_m that alters certain fields in an honest transaction T_h and assumed a Bitcoin-like POW environment. To simulate the adversary's hash power, they made building a block containing T_m easier (solving the nonce puzzle); they also injected a delay function for the honest transaction T_h to simulate any random network latency or malicious deliberate postponing (recall denial of service concerns discussed in the previous sections). In other words, the attack success would be a consequence of the adversary's success in the threats/concerns discussed in the previous subsections (A&B); deliberately delaying the honest transaction and/or controlling the network to put his/her block first.

IV. Blockchain Facilitated Attacks on All Voting Systems

We discussed in section 3.C how deploying blockchains can make coercion and/or vote buying easier. In this section, we show how the new technology provided coercers and vote buyers with new tools and facilities that they can use with all kinds of voting systems.

A. Smart Contracts and Automated Online Code Execution

1-We previously discussed how TEE execution attests, or verifying QR codes entered through an interface, can be used to trigger a smart contract, [149-151], for **automated conditional payment to the vote seller's wallet**. This can happen equally likely for any online voting system even if it does not deploy blockchains. The smart contract here provides a guarantee for sellers that they will get paid, and the anonymity property makes it easier for the buyer/coercer and sellers to hide from the authorities.

2-Blockchain online auctions, where buyers place their bids (anonymously) online, happen all the time; there is no infeasibility condition that prevents automated **online vote auctions**, [163], from happening between anonymous competitors (candidates and/or external adversaries) during election time. Different scenarios include vote sellers directly connecting to an interface site that will execute the conditional payment described above afterwards, Fig.5, or a dark web intermediary who controls many compromised vote credentials and will be paid accordingly; this may also happen (but less effective) in poll station voting to provide better timing. One could imagine, in all scenarios, the auction running till the last minutes (like race/match results bidding) where bids and votes' prices get higher with time.

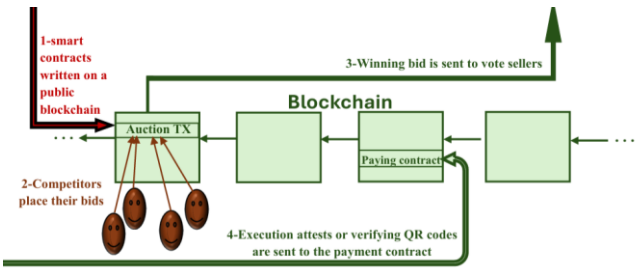


Fig. 5. A possible scenario for online vote auctions using blockchains; this is a special case modification (one threat of the more complicated combined threats in the main figure) of [163] by the same author.

B. The Immutable Public Ledger Property

The authors in [164] show how new technologies, including the presence of a public blockchain immutable ledger, can help adversaries (coercers and vote buyers) in forcing voters to comply or help voters to obtain a (probabilistic) receipt for vote selling. The paper examines the vulnerability of the important known e-voting systems with a main approach that resembles Benaloh challenge in asking the voter to act

probabilistically but here according to cryptographic randomness either produced by the voting protocol or by the blockchain. For example, post the vote or its hash/cipher on the chain, then spoil the vote if the next block number starts with zero and show me the receipt; if the voter lied and posted a wrong value at the chain, he/she might get caught with probability 50%. The paper, however, assumes the voter can post a long cryptographic bitstring value to the blockchain while in booth; i.e., the threat is more significant for online voting where the voter can copy-paste those values and have enough time to post on the chain.

V. Working Case Examples (South Korea, Voatz)

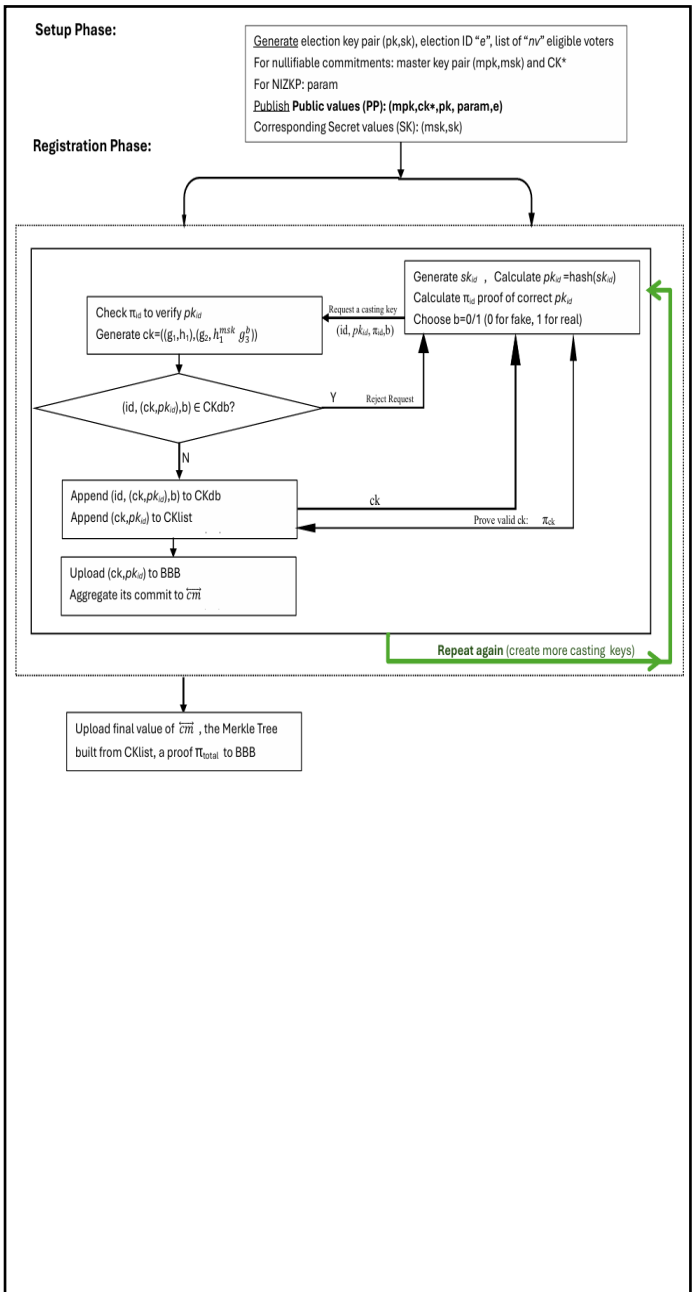
In this section, we will go through the details of some case examples and examine them against the concerns in the previous section.

A. South Korea zkVoting

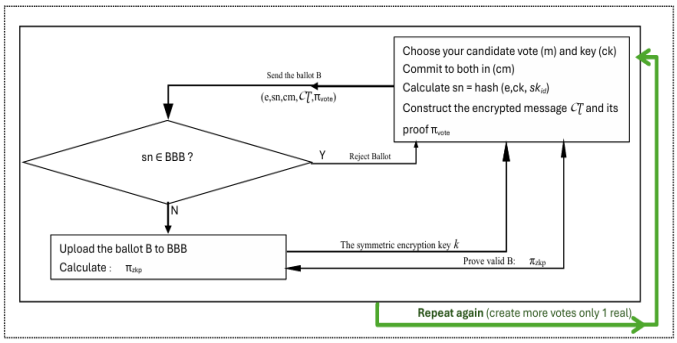
A joint work between *Zkrypto* company, Hanyang University and Kookmin University, [165, 166], won nation competitions for two consecutive years, (CES2023, etc.) [75,167], reaching this final used design.

1) Main Design

While the flowchart in Fig.6 shows the detailed steps, this section grasps the main design ideas.



Voting Phase:



Tallying Phase:

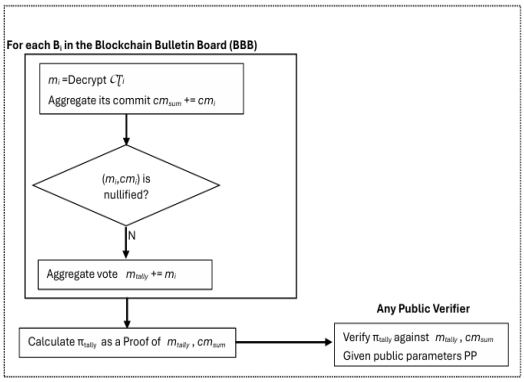


Fig. 6. The main steps of the zkVoting system, some cryptographic details regarding random numbers and relations were skipped for simplicity without affecting the core protocol design.

-Voters are provided with one real casting key (ck,1) to cast their real intended vote and as many **fake casting keys** (ck,0) to deceive coercers with fake votes; all votes (fake and real) get committed as transactions on the blockchain (assumed as the PBB, sometimes referred to as BBB in their paper adding an extra "B" for Blockchain) where only the tallier can distinguish the real vote from fake votes. To do so, the Election authority (tallier) has another public-private master key pair (m_{pk}, m_{sk}) and a **nullifiable commitment key ck*** which is used to nullify keys as fake and encrypted ballots of fake keys open to zero value when aggregating all votes.

-Voters are provided with **digital receipts** linked to their voting transactions on the blockchain using a serial number (calculated by the voter from his/her keys); the receipt includes a symmetric encryption key to decrypt the ballot (like the QR-code could be verified on any device).

-Another cryptographic proof is added to **verify the total number** of real keys equals the number of aggregated commitments after nullifying fake votes. All keys (fake & real) are stored in a list, **Cklist**, where *proof of set membership* is applied before approving the ballot; in their implementation [165] Merkle Trees are the used construct and the Merkle root is stored on the Bulletin Board BBB. Note that the election authority also keeps a more detailed secret list **CKdb** secretly to make sure every voter issues only one real casting key (in the registration phase of Fig.5, the request is rejected if the voter already has a tuple with b=1).

2) Fake Votes (Fake Credentials vs. Fake Casting Keys)

The idea of deceiving coercers with **fake credentials** was originally introduced in 2005 in *JCJ* [168] and a modification of it was implemented in *Civitas* 2008 [169]. The introduced

system, *zkVoting* [165], used fake keys instead of fake credentials for two purposes:

1. Removing trust from the registrar since the entity generating the keys is different from the entity collecting the votes; i.e., attacks can only happen if the registration application and the voting application collude together.
2. Decreasing the time complexity of the tallying phase (distinguishing the real vote from $O(n^2)$ to $O(n)$ where n is the total number of ballots fake and real) since the use of nullifiers with fake keys will make fake votes open to zero value.

3) Nullifiers and Nullifiable Commitments

Nullifiers were first introduced in *zerocash* protocol 2014, [170], which aimed to improve on zerocoin[20] (shares 3 authors with the one behind Zcoin [170/ref. MGGR13]) and eventually evolved to the privacy focused cryptocurrency ***Zcash***. What the authors called *serial numbers* at that time and used to mark the spending of a coin (i.e. to prevent double spending afterwards) was globally called *nullifiers* in further research[21]; the main design architecture of a nullifier set and a commitment tree became a standard approach in many privacy preserving chains. Similarly, *zkVoting* uses Cklist of all (ck,pk_{id}) and a Merkle Tree to provide proof of set membership[22]. Other uses of nullifiers emerged with time like validator "*unlinkability*" in Ethereum [171].

Regarding e-voting, the main concern of this paper, we trace the first usage of nullifiers for coercion resistance in a master thesis (2022) [172] and a follow up paper (2025) [173]. The equation (15) in [172/sec.4.1.2] is quite similar to how ck is calculated in Fig.5 (the registration phase); however, there are a few significant differences:

1. nullification in [172,173] happens after casting the vote only in case of coercion. In other words, the main semantic difference and/or novelty in [165] is that fake keys are determined from the beginning at the registration phase, while [172] follows the same theme of [170] that nullifiers may or may not be added according to the user spending pattern (voter coercion circumstances). As a consequence, the total number of nullifiers in [172,173] could be used by election observers as an estimate on general coercion/vote buying attempts, while the earlier generation of fake keys in [165] is not; the **coercion evidence** idea elaborated in [173] is originally due to *Caveat Coercitor* [173/ref.18] which is an 2013 advancement on JCJ (2005) and Civitas (2008). In fact, the coercion resistance strategy of *zkVoting* resembles *Decoy ballots* [173/ref.7 & sec.1.2 &Table1] in the sense that they are requested during registration (prior to voting), although differ in using a cryptographic approach nullifiable commitments instead of a game theoretic one.
2. Also, the protocols in [172,173] do not use **blockchains** and are introduced as an overlay, add-on, or "back-end" for the "front-end" of any E2E coercion resistant voting scheme; this does not prevent adding them with blockchain protocols though. In the thesis, [172] was implemented over *VoteXX* system [172/ref.17] improving on Mix and Match protocol [172/ref.39] concentrating on computing binary XORs under encryption; the overlay of [173] was illustrated on VoteXX and on JCJ [168] to only need an additional $O(n)$ at tallying time with some cryptographic tricks and concurrent processing.
3. The main strategy of [173] is to allow each voter to declare a trusted helper (named *hedgehog* strategy) who can nullify or flip votes if the voter was under great pressure by coercers and cannot do it by him/herself. It is our belief, the authors of this paper, that the whole idea collapses when considering the possibility of **a coercer/vote buyer forcing voters to be their trusted helper**; pretty expected from large-scale vote buyers in authoritarian regimes. This cannot happen in zkVoting unless the coercer attended or controlled the whole steps of the registration phase.
4. The authors in [173] put **key deniability** as a future work, while *zkVoting* proves it is satisfied in [165/Appendix B/Theorem 2.1]; put it simply, unless a coercer attended the complete registration phase to know the value of bit "*b*" sent with each request of casting key, the coercer cannot know whether a certain key "*ck*" is fake or real.

Without getting into the details of those protocols, we clarify that nullifiable commitments, or nullifiers in general, is a zero knowledge proof idea that is not necessarily bounded to blockchains although first appeared there. In fact they are one of many examples where the fast evolving rhythm of blockchain technology, including zero knowledge research, enriched e-voting research with new tools and/or ideas.

4) Security Analysis Against Mentioned Threats

1-As for malleability, the authors assumed from the beginning that their nullifiable protocol is binding, [165/sec.2.1/def.2.3]; i.e., no two ballots can be committed to the same value. Formally, for a key k (real or fake) probability of $[\text{Commit}(k,m_0;r_0) = \text{Commit}(k,m_1;r_1)]$ is negligible.

2-The execution attest and/or liquefaction of voting token attacks are not applicable here; the voter can deceive the coercer to receive the attest and pay the money by performing the registration phase from another device and nullifying the used casting key. Hence, unless monitored the registration phase to know the real key with (b=1), a rational coercer/vote buyer will realize from the beginning this is not enough as proof of compliance. Of course, this would be much easier for voters if the protocol allowed further nullifying time gap

[20] The same *Zcoin* used in the Thailand pilot 2018; although Zcash existed at that time, the secrecy of the TX destination was not a problem for an e-voting application as it is always the Election Authority (EA), while the amount (could reflect the chosen vote) may have been hidden through randomization.

[21] footnote 2 in the latest Zcash protocol specification on 15 Sep 2026 (Versionv2026.7.0-191-g0fae78 [NU6.2]).

[22] In [170], the user may spend a coin (corresponding to posting a vote using ck) by sending the nullifier and a proof π that "*I know r such that Commit$_r$(nullifier) appears as a leaf in a CRH-based Merkle tree whose root is rt*".

between the voting phase and the tallying phase; i.e., for protocols like [172,173].

3-Both JCJ and Civitas were found vulnerable by the authors of [164/sec. 3.6] if the coercer forced the voter to begin with a certain public key K_{VE} before the registration starts. The authors here assume a nullifiable commitment scheme should be hiding; in [165/Appendix A.1/Fig.3], even an adversary knowing the nullifying key ck^* and the master public-private casting keys pair, (mpk,msk,ck^*), cannot guess with probability more than ½ which keys the voter has nullified (whether a certain casting key ck is accompanied with a bit b=0 or b=1).
-Moreover, the protocol is assumed to have *indistinguishability of keys* to the commitment framework; this means that knowing two casting keys where (ck,1) is real and (ck',0) is fake, the coercer cannot identify from the commitments published on the blockchain (PBB) which key is used for each ballot (cannot guess with probability more than ½).
-However, similar to its predecessors [165/ref. 11&12], **it all depends on a safe registration phase; if attended the registration phase, the adversary can know the real casting key (ck,1) and also the secret key sk_{id} used in the voting phase to cast the ballots.** If the coercer used those values to cast a desired vote (could be from another device) and got rejected, he/she will know the voter deceived them and voted with the real key to another choice.

4-The authors stated in both presentations [75/video,165/video] that the government requested that the Election Authority (EA) generates all keys in advance to increase usability; we find real concerns regarding **Ballot Stuffing** in this case since registering in place of absent voters (and hence voting on their behalf) becomes possible. The authors talked about protecting ballots secrecy through splitting and randomization but did not discuss the threat on integrity; i.e., considering that voters can avoid such threat by participating in election, or considering EA to constitute a quorum of trustees [174/Quorum for decision] under threshold cryptography (cannot collude beyond the threshold). In any case, if another Merkle tree (IDlist) was constructed from participating (invoked the registration phase with b=1) voters'ids with its root postedon the blockchain, absent voters could be encouraged to check that their ids are not included (non-inclusion proofs are supported by Merkle trees); election observers and public verifiers could also check a random sample. In this case the *immutability* of the public ledger can serve as an added value to secure the consistency between IDlist and posted Ballots. However, in this dispute case of the voter against EA, there is no way of telling (probability = ½) who is lying; still, the IDlist can provide an earlier alarm if there was a time gap between the registration phase and the voting phase, here and in the following cases of voter penetration as well.

5-Although related to online voting in general, not specific to blockchain deployment (could happen with any PBB), a fake voting application accompanied by a fake registrar can deceive the voter with a nullified digital receipt. Assuming the adversary *A* is an eligible voter, **with a fake application *A* can take the voter choice, cast a corresponding vote using *A*'s own fake keys and sends the voter the digital receipt**; this will kind of switch the places between the voter and the adversary making the system treat the voter as a possible adversary who has no right to distinguish whether the vote is real or fake. This can work only if the registrar was also a malicious rogue application (or if there is only one application for both phases in the real implementation as the demo in [165/video]) to send the victimized voter fraud credentials, some of which belong to the coercer; the voter will find sn=hash(e,ck,sk_{id}) and other cryptographic verifying values seeming to be correct in this case. This issue was questioned by one of the audiences in [165/video] referring to Ron Rivest previously mentioned concerns on online voting; the answer of checking the "*sn*" value is not quite sufficient here since the fake application can copy to the voter exactly all the adversary's correct keys except which key is fake. As for comparing with [173], we believe this attack is worse than the above-mentioned one where the coercer being the trusted helper since the voter is not aware of the problem here; like the previous threat, putting the original id of registered voters into a Merkle tree would make the attack detectable by the voter if his/her id was not included in IDlist.

6-If the attacker *A* can deceive the servers as well to register in place of the voter (man-in-the-middle attack) then the IDlist will not help here. *A* can register all the voter keys as fake and keep the real key to vote with *A*'s choice; the voter will find all his/her votes with their *sn* values correct (cannot distinguish whether they are fake or real) and also will find a membership proof for his/her original id in IDlist. However, although less probable and against the design goals, if in actual implementation EA previously generates only one fake key for each voter, *A* can only switch them for the voter (but not vote in his/her place).

7-A similar threat to the above, although less severe, is for an **eavesdropper** adversary who captures **the registration phase** keys (the case of leaked keys in [173/Table2]) then quickly votes with the true key before the voter. In fact, [173] mentions an axiom by [173/ ref. 21&22] that "*if the adversary has/knows/is everything that the voter has/knows/is, the voting system cannot distinguish the voter from the adversary and therefore cannot provide a mechanism to register the true voter's vote and not the adversary's*". However, the voter (if not coerced) can report to the authorities that the system rejected his/her vote for being already used when it is not; i.e., since *sn* depends solely on the registration phase values, the voter will be able to find it on BBB. Hence, this attack is detectable and could be resolved by following [173] in adding a time window between voting and tallying for such complaints.

8-Finally, there might be some **scalability** issues for country level elections where the paper implementation holds 2^{16} ballots ≈ 65,000 votes (the Merkle tree holding all vote hashes including fake ones). The author in the presentation [75/video] says the system handled over 100,000 votes since launched on 30 Nov 2024 with peak participation of 15,043 votes on 27 Dec 2024; for comparison basis, the Merkle tree of Zcash (2014), [170], was of 2^{64} (> 10^{18}) nodes. The scalability vs. trust issue was mentioned also in [166/sec. 3.3] who questions the public blockchain used in the actual implementation since the paper used a local Ethereum Ganache environment; if all used Ganache nodes and servers

belong to the government, then the design loses most important public blockchain features. Recalling the quorum of trustees mentioned in [174], we assume the standard threshold cryptography techniques between the government and other trusted third parties. In addition, like most existing e-voting systems that deploy blockchains, aggregated partial results and the final result are always posted on the public blockchain; recall the snapshots in Agora [20] and zk-rollups in Vocdoni [21]. For *zkVoting*, it will not add extra cost to post the Merkle root on Ethereum L1, and also periodical partial aggregations which helps voters to check not just their *sn* value is included but within the correct time interval they voted in; this makes it harder for fake applications to repeat the scenario in "4" for many voters using the same fake ballot.

5) Concluding Remarks

The idea of fake credentials remained in literature for so long, since 2005, before being used in actual elections. The given protocol should take the credit for an actual running implementation that scales up to 100,000 ballots if claims in [75/video] are true; we believe any performance efficiency issues could be improved following the original source of the nullifiers idea (Zcash [170]). We identified some threats and/or security vulnerabilities, some of which are common to all online voting systems; we believe most of them can be solved by anchoring more pieces of data (a cryptographic fingerprint of all participating voter ids, whether inside CKlist or in a separate Merkle root). Like all nullifiers and/or fake credential approaches, attending the whole registration phase (knowing all its keys) remains an undefeatable threat; when suggesting a time window between voting and tallying for an extra nullification phase, we should be aware that coercers can attend it as well.

B. Voatz

In this section we try to summarize and analyze the available material around Voatz with all the criticism that led to every US state taking a separate independent decision about it; ex. [98].

1) Overview

Started in 2015 by an award-winning entrepreneur, conducted pilots in some USA states in 2018/2019 and expanded beyond many USA states to general elections in Mexico and 51 Ontario municipalities in Canada by now (https://voatz.com/blog/), Voatz is an online mobile voting system that is mostly used by remote voters. Exploring the official site (https://voatz.com/whitepapers/), earlier whitepapers [175,176] start by emphasizing the importance of providing solutions for persisting problems like less participation by remote voters and/or losing their votes by post office mail if arrived after deadline (recall footnote 6); as complained by [96,97] not very much technical and/or cryptographic details are included in them ([175] clearly states "*This paper is meant for a curious, non-technical reader*"). Mostly discussed voting steps are the authentication methods and the blockchain usage.

Voatz replaces the traditional public bulletin board (PBB) with a semi-private permissioned blockchain and public post-election tabulated ballots list [97/blockchain servers, 175,176]; a whitepaper by NASS (*National Association of Secretaries of State*) in 2025 [177] about audits via blockchain, cites [178] (2018) as the original peer-reviewed proposal of the blockchain usage idea. Another 2023 paper, [179/pages 29,30], is also featured in the official site for including a section about Voatz; the paper discusses the degree of decentralization in e-voting systems and considers the blockchain to be the only decentralized component in Voatz design, while the rest parts/phases take a centralized approach.

2) Main Voting Steps

The Voting steps, as illustrated in Fig.7 taken from [175], are:

1. **Voter Authentication** [175/KYV]: a voter is identified through a photo of ID/passport and a short interactive video to guarantee liveness, in addition to the smartphone/tablet authentication; i.e., each voter must have a 1-1 bond with a device and is required to re-authenticate whenever the application goes to the background. An anonymous voting ID is assigned to each voter which is a function of all the authentication parameters (including the device identifying information); we believe only Voatz, of all governmental level online voting systems, uses this authentication methodology.
2. **Voting through the blockchain** [175/Table2]: when the election starts, each jurisdiction assigns enough credit to eligible voters (to the anonymous voting ID) that is equal to number of choices. The voter choice for a certain candidate A is implemented as a transaction transferring credit from the voter to the candidate; in addition to cryptographic checks, the transaction is validated only (the voter account is debited and the candidate account is credited) if the choice is valid for this jurisdiction.
3. **Spoiling a Ballot**: a voter can spoil an old vote and vote again but must be through the same device.
4. **Ballot Receipt:** a voter receives a confirmation message when the ballot is validated and stored in the blockchain; the receipt contains the choices and the anonymized voter ID.
5. **Audits:** post-election audits are possible for voters, jurisdiction and independent viewers; this is through what could be called a PBB as in Fig.6 or a *Tabulated Ballot* (list) as called in Fig.7, where Fig.(8-a) is taken from [175] (2019) and Fig.(8-b) is taken from [177] (2025).

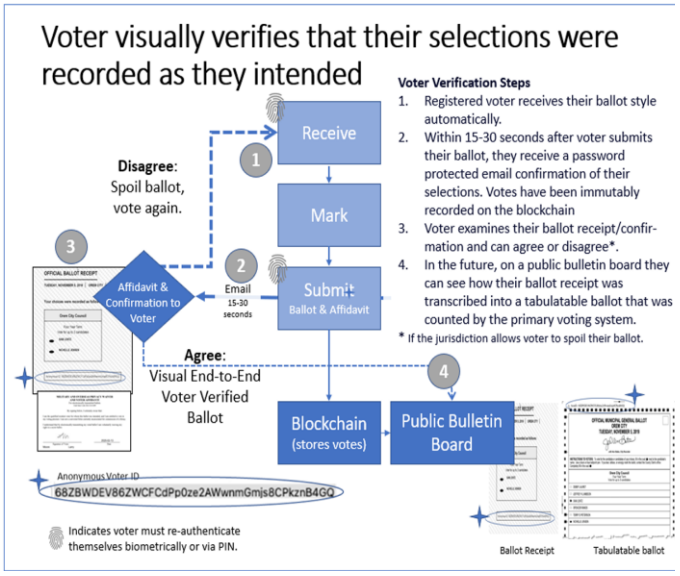


Fig. 7. Voter and verification steps in Voatz as shown in [175/Fig.2]

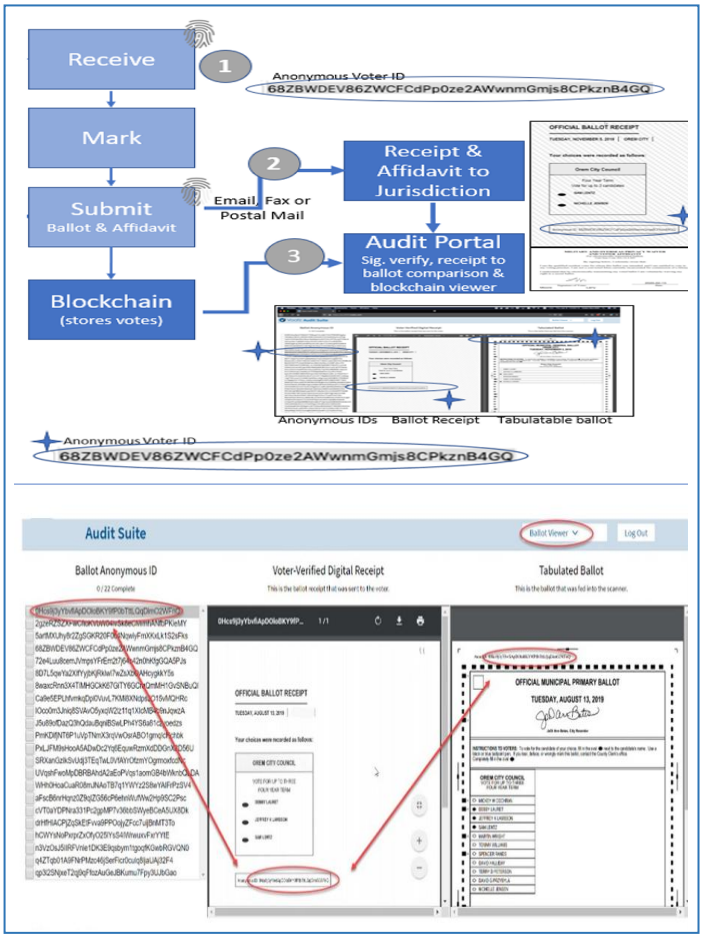


Fig. 8. Visual post-election audit procedure.

3) Known Details About Voatz Design

A main design pattern in Voatz is contracting with other companies for certain phases and/or services. Examples include:

1. Integrating (https://www.zimperium.com/) for **mobile threats defense** in USA. If the voter smartphone is found to be compromised, the device is prevented from voting; [180] shows some threat mitigations examples that happened during the pilots.
2. Deploying the **five-layer defense** strategy applied for remote voting in Mexico 2024 election using Amazon AWS (Amazon) [181].
3. The services provided by AWS for Mexico include **image recognition and video analysis** for authentication, while (https://www.jumio.com/) was used in USA elections since 2019. According to [96/footnote 8] *Jumio* has disclosed then that it uses a third party, *Facetec*, to help in analyzing the video selfies.

As for **the blockchain design**, the main concern of this paper, released white papers [175,176] do not reveal much about the blockchain design except a FAQ table from which, among other resources like [96,97], we find the following:

1. Voatz uses a semi-private permissioned blockchain that consists of 32 decentralized storage servers (split equally between Amazon AWS and Microsoft's Azure in USA, while all Mexico servers are from AWS [181,182]; again, commercial companies). The blockchain servers run the IBM Hyperledger Fabric protocol on Linux and apply Practical Byzantine Fault Tolerance (PBFT) for consensus.
2. Anonymous ballot choices are directly stored in the blockchain in USA and all other elections run by Voatz; except for the Mexican elections, [182/sec. 2.5], where only the anonymous IDs are stored in it.
3. Since the blockchain is semi-private, Voatz still uses some form of a Public Bulletin Board (Fig.6&7) to store the post-election public auditable data; this public PBB is sometimes called *Tabulated Ballots list.*
4. Although not stated explicitly in white papers, we believe nullifiers might be used to spoil old votes in USA states. However, spoiling votes was not allowed on the Mexican elections and any attempt of repeated voting is detected as a replay attack; this aligns with the design choice of storing only anonymized IDs in the blockchain as there is a 1:1 correspondence with the number of stored ballots.
5. It is important to note that Voatz does not post any cryptographic checksums on any public permissionless blockchains.

For **cryptographic details**, we know the following:

1. According to [176/Appendix A: Communication & Network Security], all data transmitted over the network is encrypted using NIST approved AES-GCM 256 bits key (with RSA key agreement with SHA, SHA256, and SHA384); also, PKI (Public-private Key pair Infrastructure) is established between the voter device and the server. The PKI key generation and selection was one of the issues raised by MIT [96/sec. 4.1.1]; the researchers claimed that both sides generate 100 keys and the 57th key is always selected. However, Voatz replied [183/sec. 2.3] that this was in an earlier test version, but the version used in the election pilots makes a random selection.

2. The technical report released about the Mexican elections [182/sec. 2] reveals the use of *Paillier* additively homomorphic encryption to encrypt ballots; i.e., encrypted ballots are multiplied then decrypted and no mix-nets are used. Also, a Non-Interactive Zero Knowledge Proof (*NIZKP*) is added to each vote online; ballots are signed with added randomness through *blind signature* that uses different prime numbers (p & q) than those used by *Paillier*; finally, *Shamir secret sharing* threshold scheme (*3-out-of-5*) was used to split the election authority key into 5 shards that can be reassembled using a committee of 3 only. However, we could not trace any document stating that those cryptographic details are applied to other elections conducted by Voatz whether in USA or Canada.

The rest available cryptographic details are better discussed one by one through the researchers' concerns.

4) Concerns raised by Researchers in the years 2019-2021

In addition to general concerns on online and/or blockchain-based voting, there were two research groups that analyzed Voatz in detail; one from MIT [96] and one from University of South Carolina [97]. The MIT group gained more publicity because they published their work in USENIX and because they were accused of hacking Voatz servers[23], while [97] remained as a preprint. This section discusses all concerns and/or vulnerabilities introduced by [96,97] along with Voatz reply to them [183]; since those debates took place in 2019/2020 and some replies concentrated on the pilot experiments only, we will try to discuss the general case (all eligible remote voters, not just a sampled group) and do the necessary follow up from the most updated site.

a) Certification

Voatz repeats in many places of [175,176] that it uses NIST (*National Institute of Science & Technology*) approved algorithms and applications. However, the authors in [97/certification] discussed that Voatz is not certified to the EAC (*Election Assistance Commission*), [184], standards and that EAC did not include standards for internet voting at that time. The authors did not deny though that Voatz seeks other security audit certificates from its affiliated partners (https://voatz.com/partners-and-affiliates/).

A significant audit was conducted by *Trail of Bits*, [185], in Mar 2020 to serve as a trusted third-party referee between Voatz and MIT researchers [96]. *Trail of Bits* reported 48 threats and classified 16 of them as severe; Voatz responded, [186], by resolving 20-22 of them while considering 17 as false positives and 9 as irrelevant.

After the fixes, July 2020, Voatz successfully completed comprehensive system testing with a Federally certified VSTL (*Voting Systems Test Laboratory*), [187] as a phase1 requirement for EAC certification; the certificate says that Voatz "*meets the applicable requirements set forth for voting systems in the U.S. Election Assistance Commission (EAC) 2015 Voluntary Voting System Guidelines (VVSG), Version 1.1,* ***with clarifications or exceptions noted in Section 4.0***", unfortunately we found no trace of the complete report to read section 4 of it. Also, those guidelines have been updated in 2021 to VVSG 2.0 which is mandatory for new federal VSTL accreditations and certifications since late 2023, and the official site (https://voatz.com/audits/) does not show that Voatz is tested under the new ones.

Still, the site shows 9 more independent significant security audits Voatz has conducted since then (mostly with its affiliated partners); the latest two (2025) are application penetration testing for Canada voters and an audit on the organization's ISMS (*Information Security Management System*) compliance (ISO/IEC 27001:2022). However, for the year 2024, we ought to clarify that [182] is not a "*peer-reviewed paper*" as stated in the site; it is published in IEEE Technical Blockchain Community which does not perform a peer-review process. Although the approval of the Mexican election commission could be considered as a credible security audit, we find it misleading to include [182] as one.

b) Hiding Design Details & Control of Commercial Vendors

This was mentioned earlier under control of **blockchain server owners**; also, under transparency regarding the details of the exact **cryptography** used [96] and the exact fields stored on the **blockchain** and under what form.

Another concern that was highly raised by [96, 97/voter identification and authentication] is the methodology *Jumio* uses for **image and video analysis**; the exact machine learning and video analysis techniques used were hidden and the code is not open sourced. They questioned the threat of possible manipulation by the company; i.e., blocking certain voters on the claim that they failed the authentication check and/or accepting fake voters. Voatz, sufficed with replying, [183], that an additional human identification step is added to detect any identification errors whether malicious or not.

On the threat of **controlling the blockchain** by one of the vendor companies (Amazon & Microsoft), Voatz replied that it does not use a POW consensus to be subject to 51% attack; since the blockchain used is permissioned (PBFT consensus), companies cannot just add more nodes (servers) without permission. Although Voatz reply is right that 51% attack mistakenly mentioned by MIT in [96] is not possible, we remind that PBFT is resilient only with < 1/3 malicious nodes; i.e., their final ratio estimation of 1/3 the total power remains accurate though. Many of the questions raised by [97/blockchain servers] in 2019 are answered by now, except the concern about the possibility of a software bug in the consensus/operating and whether it would be repeated on half or all the servers. In fact, this complies with the threat we alarm about here and mentioned in footnote 14 and the section on controlling permissioned chains; PBFT can handle only <1/3 malicious nodes and each vendor controls 50% of the nodes. Had it hypothetically happened that one of those 2

[23] We are not sure of whether Voatz proceeded with the accusations to report MIT researchers to the FBI or not, but the asserted fact is that there was a previous 2019 investigation with students from Michigan University who attempted to break into Voatz servers during 2018 pilots (as part of a course project activity on election cyber security); (https://www.michigandaily.com/section/news-briefs/university-michigan-students-implicated-potential-voting-app-hack, https://edition.cnn.com/2019/10/04/politics/fbi-voting-app-hack-investigation/index.html)

vendors declared a certain bias in the US presidential elections, we doubt that Voatz will be able to convince cryptographers and blockchain experts that it is still Okay to let that company's servers constitute 50% of the blockchain nodes.

c) Altering/Discarding/Blocking the Vote

1. THROUGH MAN-IN-THE-MIDDLE ATTACK (MITM)

-Voatz stated in [176/Appendix A] that it uses *certificate pinning* to prevent/detect man-in-the-middle attacks; i.e., the application checks the server's cryptographic certificate against a local copy it has to make sure an adversary did not intercept in the middle and altered information sent between them.

-However, [96/sec. 4.1.1] claims that there is no authentication of the communication session keys (digital signatures) by the application, beyond the encapsulating TLS certificates which makes no protection against active MITM attacks. The researchers concluded that because it made it easier for them to direct the ballots to their own server (instead of Voatz servers) by simply replacing all required URLs in-app by their own and follow the protocol.

-Voatz on the other hand, [183/sec. 2.11/c)], denies those claims and describes them as "*blatantly incorrect*" asserting that public key authentication is performed as part of the initial voter onboarding. To explain the URLs easy replacement, Voatz accused the researchers of attempting to break into the official servers in Dec 2019 and denies their claim that Voatz servers were down during their experiment in Jan 2020.

2. THROUGH MALWARE OR FAKE APPLICATION

-As an outsider looking at the events to analyze it, if Voatz is right about the accusations and the researchers did not replace the URLs letting their experimental ballots go to the official Voatz servers, this means that the servers did not raise an alarm when receiving those fake ballots.

-If the researchers conducted their experiment in Dec 2019 (i.e., not in Jan 2020 as they claim) and the system backend interpreted the data they sent to the servers as a breaking attempt, this is not good enough for the voter side. Such a possibility means that if any malware on the voter device went undetected and tried to modify the voter choice using the same techniques used in [96] to modify the application original workflow, although the servers will be alarmed the voter is not.

-An analogous scenario is presented in [96/sec. 5.1.2/Stealth UI Modification Attack]; this was preceded by demonstrating how disabling the *Zimperium* malware defense SDK was possible in a few lines of code [96/Fig.7]. In their report, [185/p.20&60], Trail of Bits also pointed out that Zimperium can be circumvented (medium severity: TOB-VOATZ-029); Voatz on the other hand, [186], considered the threat a false positive.

-A third threat (in the general deployment if not in the pilot) is the possibility of a voter being deceived to download a fake malicious impersonating the Voatz application. The fake application could show this voter similar screens to the original ones.

-A fourth threat introduced in [96/sec. 5.3] when discussing side channel attacks is that biased ISPs (internet Service Providers) can block votes against their preference. If this is feasible, it is a threat even without side channel attacks; could prevent ballots from certain locations and/or IP addresses if they have known dominating preferences for example.

In all these cases, the only defense [183/sec. 2.11/b)] is the voter's email receipt and its corresponding anonymous copy for officials and auditors; this is enough defense in a pilot experience but is it enough in the general case when applied to all remote voters. Practically, during general application, how many voters could be deceived by a fake email receipt or would not check it at all; how many would also proceed to search for their vote in the final tabulated list. Furthermore, the third scenario of a fake application could be combined with a Bulletin Board *Clash attack* using one true vote for all voters with the same choice; the fake app can show them all the same anonymous ID Voatz provided the adversary for his/her only one counted vote.

Hence, unless there exist extra defenses that we and [96] missed, the system's integrity heavily depends on using a strong random sampling auditing mechanism to check errors and can benefit from a similar RLA (Risk Limiting Audit) approach as that in [188] (recall footnote 6). The adaptation we suggest is to perform dual regular mail voting for all voters, as in [188], then check a sample of received post office mail votes using RLA techniques. Only when RLA detects a significant error, online results are aborted and the dual mail votes are counted. Conducting RLAs on a volunteered sample of remote voters and another sample of observers (without a complete dual vote) is susceptible to manipulation as the sample will not be statistically random.

d) Privacy Concerns

Although MIT researchers acknowledge [96/sec. 4.4] that "*Voatz employs a number of obfuscation techniques, leverages a third-party virus scanning service, and uses an on-device encrypted database to protect locally stored sensitive data*", they and [97] raised many privacy concerns and elaborated on their implications:

1. Privacy concerns stemmed from all the collected authentication data, binding each voter election ID to a fixed smartphone/tablet[24] device; [97] stressed on whether keeping the device ID after anonymization would leak some information about the identity of the voter, and the choice of the company handling biometric authentication. Both [96,97] questioned the guarantees that third parties will delete voters authentication data (copy of ID/passport, device pin, short video, …); MIT researchers also warned, [96/sec. 5.4] from collecting information about remote military deployments through voting. Voatz replied, [183], that all data transmissions are encrypted every step

[24]The authors in [97] doubted whether tablets are included, but the whitepaper [175] repeatedly writes smartphone/tablet which means that tablets are allowed.

of the way and get deleted in a maximum of 24 hours.

2. The possibility of side channel attacks on the transmitted vote over the network (before anonymization) was discussed in [96/sec. 5.3] and featured as a main example in their presentation [96/video]. Voatz replied, [183/sec. 2.9] that the researchers used skewed flawed data to support their claim while different candidate choices are formed in the code to result in an almost balanced load. On the other hand, [96/sec.6] claims that the authors have alerted the U.S. Department of Homeland Security (DHS), and that "*they appeared to confirm the existence of the side channel vulnerability, and the PIN entropy issues*".
3. There were also concerns about being able to define the voter through the time of casting since a permissioned blockchain keeps the order and timing of transactions and there were not enough details on when (at any stage) mix-nets were used to shuffle the votes if used at all.

In addition to the defense replies mentioned above, Voatz simply reminded of the fact that remote voters already sign a waiver on their vote privacy rights when voting through regular mail. In their exact words [175/a note on absolute anonymity] "*However, given that every state needs to know who has voted (a requirement called "voter credit") and given that nearly every state requires the remote voter to waive their right to anonymity,* ***Voatz can make it extremely difficult, but not always impossible, for the election department staff to pair a voter's ballot with their identity***". However, some of those threats exceed the vote privacy to the voter personal data and could lead to discarding/blocking the vote if the adversary has the power to do so (ex. ISPs mentioned in the previous section).

5) Concluding Remarks on Voatz

We find most of the concerns raised about Voatz are not actually about its blockchain deployment but come from a common pattern of vulnerabilities any internet voting system might be susceptible to. At the end, it is a strategic decision for each country and/or state, which we recommend must be combined with strong auditing and sampling techniques (Risk Limiting Audits). However, we believe transparency is a major justified concern; we only learned some insight on the cryptographic details of the Mexican implementation through the technical report in [182] and we still do not know if they are the same in the versions used for USA and Canada.

Regarding blockchain usage, when it is semi-private permissioned although the chain is controlled by those who have the permissioned access rights on them, concerns remain valid for the servers' owners (especially if they are commercial large-scale companies). On the other hand, the permissioned chain paradigm makes it possible to extend the permission rights beyond the governmental Election Authority (EA) and include independent observers; since PBFT consensus is resilient for up to f malicious nodes vs *2f+1* honest, one could imagine accepting *f+1* servers from independent observers for each *2f* government owned/controlled servers.

VI. Summary & Conclusions

In this paper we tried to gather and organize the existing material on the usage of blockchain technology in e-voting systems to be able to reach some conclusions.

We found two main approaches; the first is to use a private or semi-private permissioned blockchain for storing encrypted ballots, **partially replacing the usual Bulletin Board**, while the second is to **post accumulated cryptographic checksums** of election data on one of the famous public permissionless blockchains (i.e., Bitcoin or Ethereum L1). Some systems combine both approaches (ex. Agora, Vocdoni); while some earlier research suggested posting all ballots on a public permissionless blockchain, this design tend to be practical only for medium to small-scale elections (or pilot studies for O(1000) voters). Table II summarizes example companies mentioned through the paper.

TABLE II. Companies that deploy blockchains in voting

Company	***Ballot Storage***	***Cryptographic Checksums***	***Countries Using it***
Agora [20]	Customized Permissioned Chain (*SkipChain*)	Periodic snapshots Bitcoin (Catena scheme) Stored in *Cotena* log	Sierra Leone Pilot (2018)
e-Vox	Ethereum	NONE	Ukraine pilot Kiev (2016)
Luxoft	Permissioned Chain (Hyperledger Fabric)	NONE	Switzerland Zug pilot (2018)
Opn.vote [22]	Inter Planetary File System (IPFS) + metadata on Ethereum	Meta data on Ethereum smart contract	Germany pilot (2025)
SimpleProof [85,120]	NONE	Timestamped fingerprints of partial results (Merkle root of hashes) Bitcoin (Catena scheme)	-Guatemala 2nd round presidential elections (2023) by civil society. -Some USA states (officially)
Smartmatic (TIVI) [32]	Digital time stamping for every vote	Cryptographic fingerprints are for every vote (not cumulative)	-Utah trial 2016 [189] -Norway trial 2018 [31] -Germany social elections 2023 [190].
STS (Romanian Special Telecommunications Service)	Permissioned Local Chain deploying Ethereum Geth	EBSI (European Blockchain Services Infrastructure)	Romania for local & European parliament elections
Vocdoni	BFT private permissioned chain (*Vochain*)	NONE	-The opposition in Belarus -Omnium Cultural organization in Barcelona
Vocdoni evolution *(DAVINCI)*	Ethereum L2 ZK-rollup	ZK-SNARKs on Ethereum L1	Has not been used in real elections yet
Voatz [175,176,182]	Permissioned Chain (Hyperledger Fabric)	NONE	Some USA states, some Canada municipalities and Mexico 2024 elections
Zk-Voting [75,165]	Permissioned (Ethereum local Ganache chain)	NONE	South Korea public institutions and local referenda

We collected most country level conducted pilots for blockchain-based e-voting to find out that they flooded the globe during the years 2016-2019 to settle down with a few real steady deployments in the years to follow. Posting cryptographic fingerprints on Bitcoin succeeded in Guatemala and some USA states, while blockchain bulletin boards are used in Russia, Mexico, and some USA states.

Looking through all discovered and/or literature discussed vulnerabilities from deploying blockchains in e-voting systems, we find that it is **a country's strategic decision that depends on many factors**. From election organizers point, it is the effect on election results they care about (read the quote from political *philanthropist Bradley Tusk* in [96/sec.6/conclusion]); for this argument, we recommend **accompanying such systems with the use of strong RLA technique** (ex. [188] ideas could be applied to Voatz too). Also, sometime the cost savings and increase in participation when combined with online voting significantly matter [176, footnote 16] to the state/country's decision. In all cases, **we find posting cryptographic fingerprints of aggregated partial results to be always an added value with no added threats**, if enough guarantees (auditors/observers careful monitoring) were provided to make sure of hashing the true values; we also recommend **entering the subtotal numbers better than scanning tally sheets**, where the hash value could differ according to the paper position on the scanner and/or its resolution. Major countries who fear blockchain control may invest on having a ratio of the blockchain nodes and should assess a few large-scale commercial companies (the case of Voatz) as a higher severity threat.

In addition, this paper acknowledges **the added value of blockchain technology to research on cryptography** and Zero Knowledge Proofs (ZKPs) in specific; many e-voting systems benefited from *nullifiers* originally developed for Zcash and many more systems use (with or without blockchains) NIZKPs that were originally developed for a blockchain/cryptocurrency related purpose. The paper also points out **new coercion and vote buying innovative techniques that became possible through the existence of blockchains'** public immutable ledger and/or smart contracts online coding capabilities; it is quite feasible to conduct online vote auctions, to automatically vote with the adversary's choice in some systems, and for the coercers to force a coerced voter to post some intermediate revealing values on a public blockchain. Again, all those threats do not depend on deploying blockchains in the used e-voting system.

Finally, we detailed two case study systems in this paper. The first is **zk-voting with nullifiable commitments** in South Korea and compared it with another system that uses nullifiers without blockchains. Then, we got into the details of **Voatz** with all the debates and criticism that happened around it in USA (2019-2021) and discussed its 2024 Mexican modified system as well.

We hope this paper provides added value for policy makers on their decisions on whether to use blockchains on their e-voting systems after reading all the relevant facts; also, to become aware of possible new attacks even if they did not use blockchains. As for academic researchers, whether on e-voting or blockchain areas, we hope the paper provided a thorough survey and accurate analysis/concluding remarks that constitutes a starting point incentivizing new research. We as authors became incentivized to perform deeper analysis and comparisons of Ethereum based e-voting systems (*opn.vote* and *Vocdoni*) as possible future work.

RESPONSIBLE DISCLOSURE

The author has notified most involved parties in the case studies section with a copy of the paper draft referring to the exact section their work is discussed in case they have any comments. This includes the sites of zk-voting [75], the site in [76] discussing it from South Korea, the authors of the master thesis from Canada about the use of nullifiers in coercion resistance and its follow-up paper [172,173], the authors of the MIT paper about Voatz [96] and finally Voatz on X formerly twitter.

AI USAGE

In general, AI helped in searching for more accurate resources to refine the mixed-up and/or sometimes misleading information about older pilots; also, very recent ones (ex. *opn.vote* paper from Germany appeared during the writing phase, EVoteID'26 and would have been missed without AI).

Then through Google Deep review, the technical report about Romania usage of EBSI [38] was discovered (the only other citations encountered was [37] through chainlink). Finally, this paper was revised through Gemini, Copilot and then Google Deep review before submission.

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